

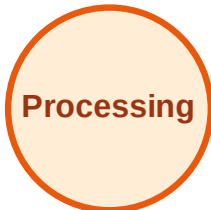
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



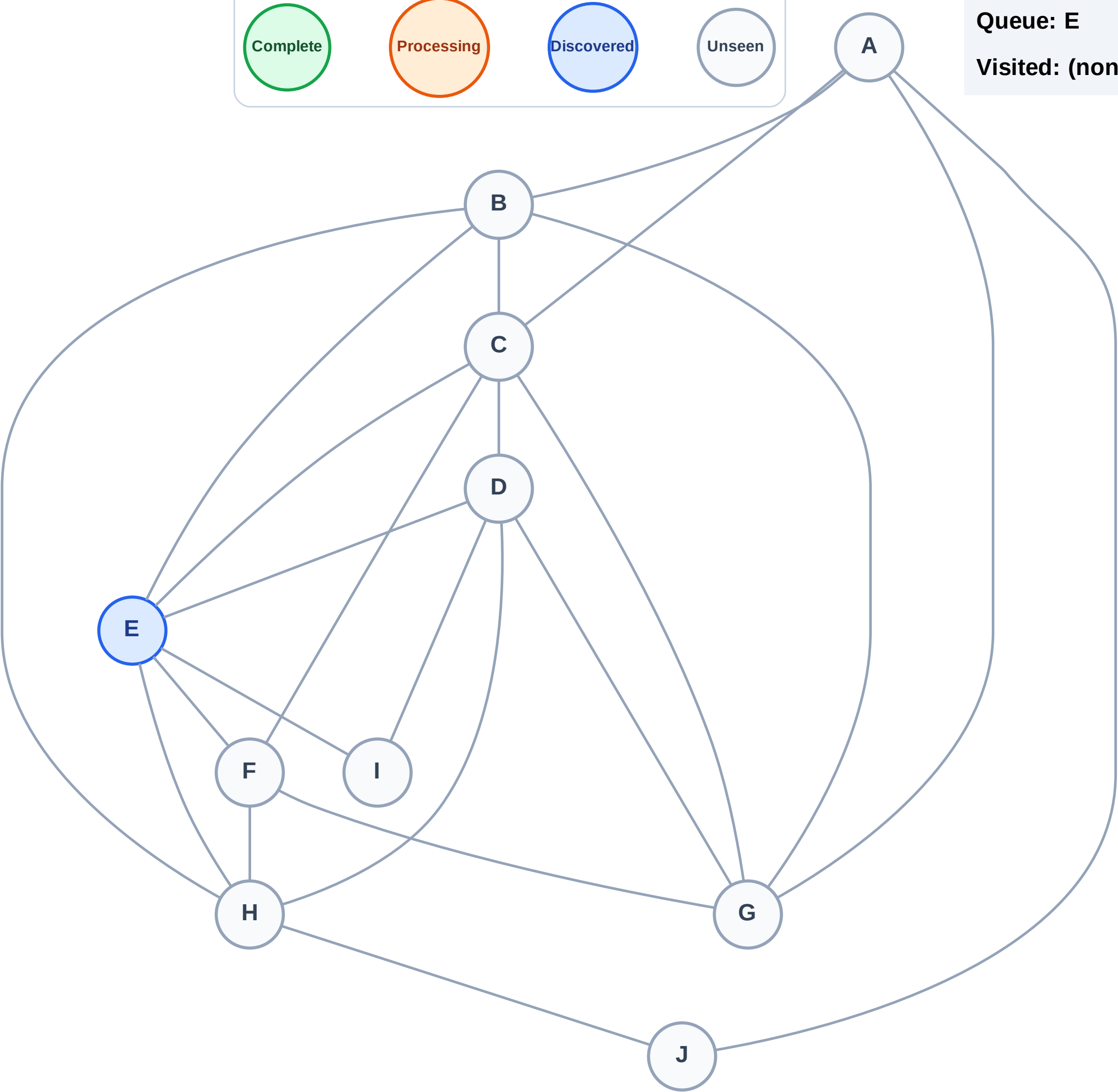
Discovered



Unseen

Queue: E

Visited: (none)



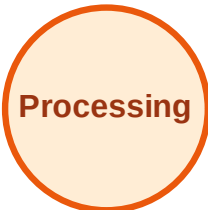
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



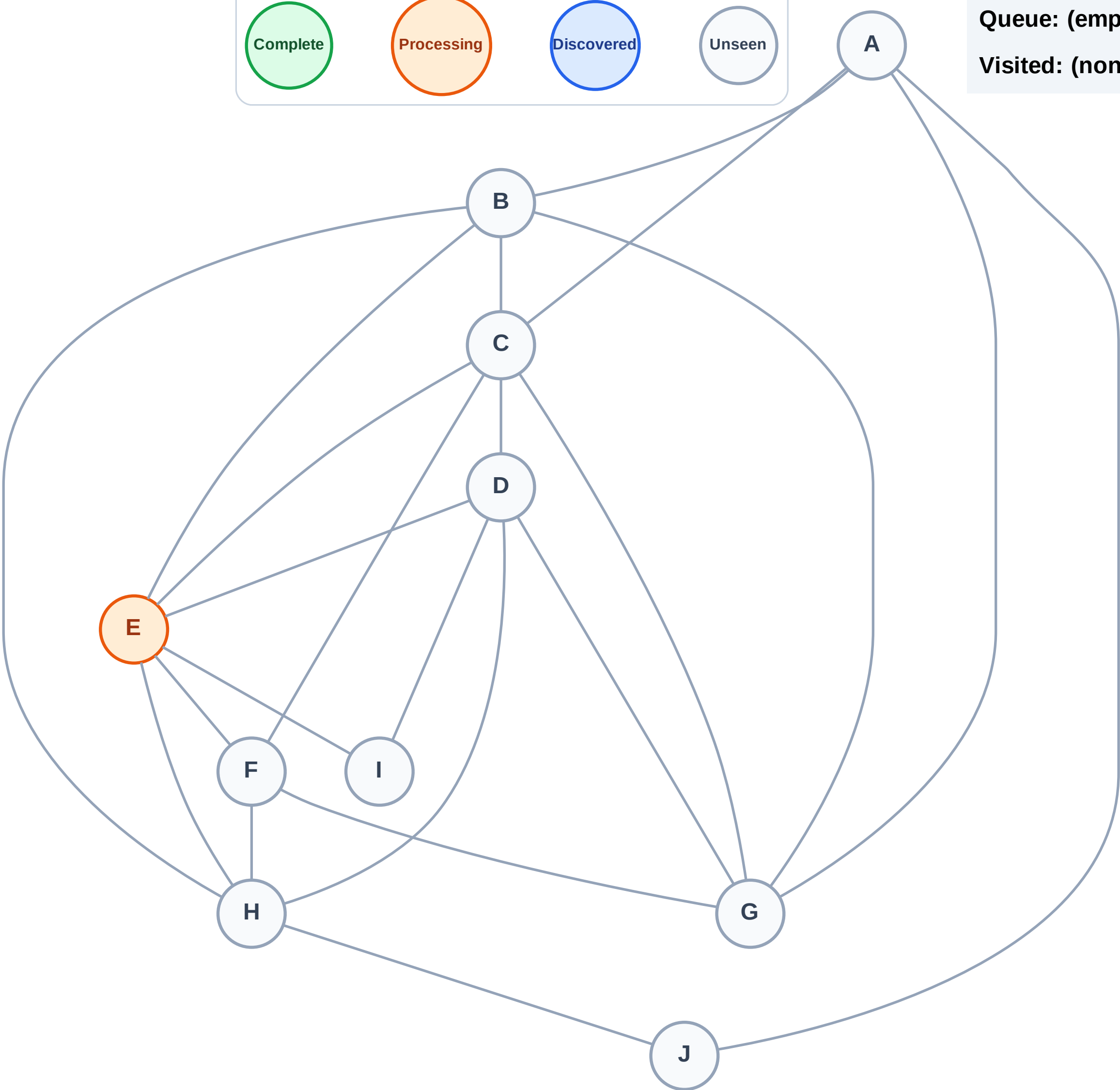
Discovered



Unseen

Queue: (empty)

Visited: (none)



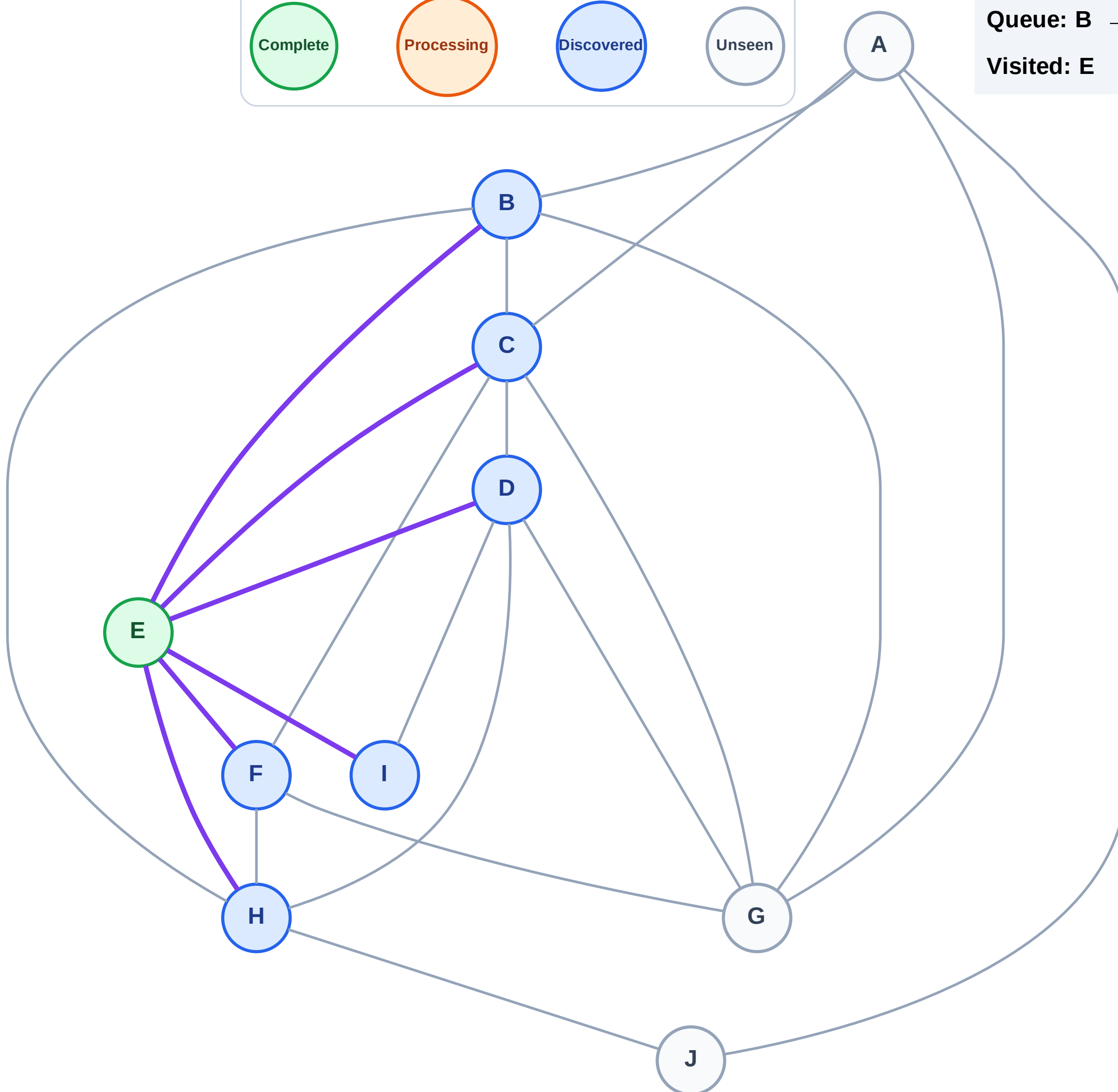
BFS Traversal

Step 3: Discover B, C, D, F, H, I from E

Legend



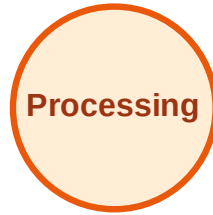
Queue: B → C → D → F → H → I
Visited: E



BFS Traversal

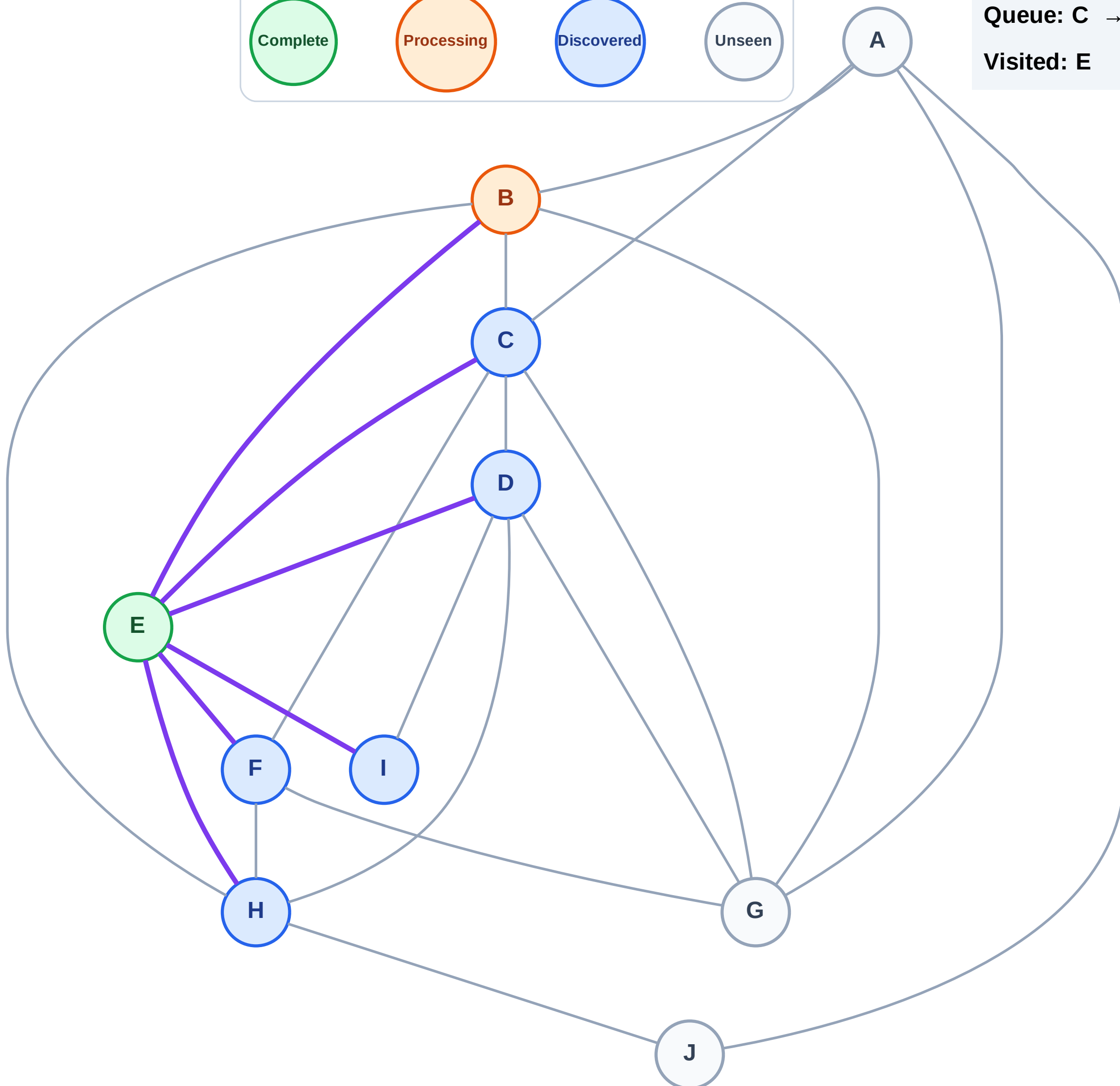
Step 4: Process node B

Legend



Queue: C → D → F → H → I

Visited: E



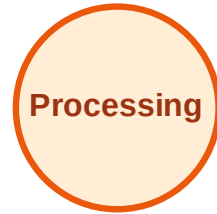
BFS Traversal

Step 5: Discover A, G from B

Legend



Complete



Processing



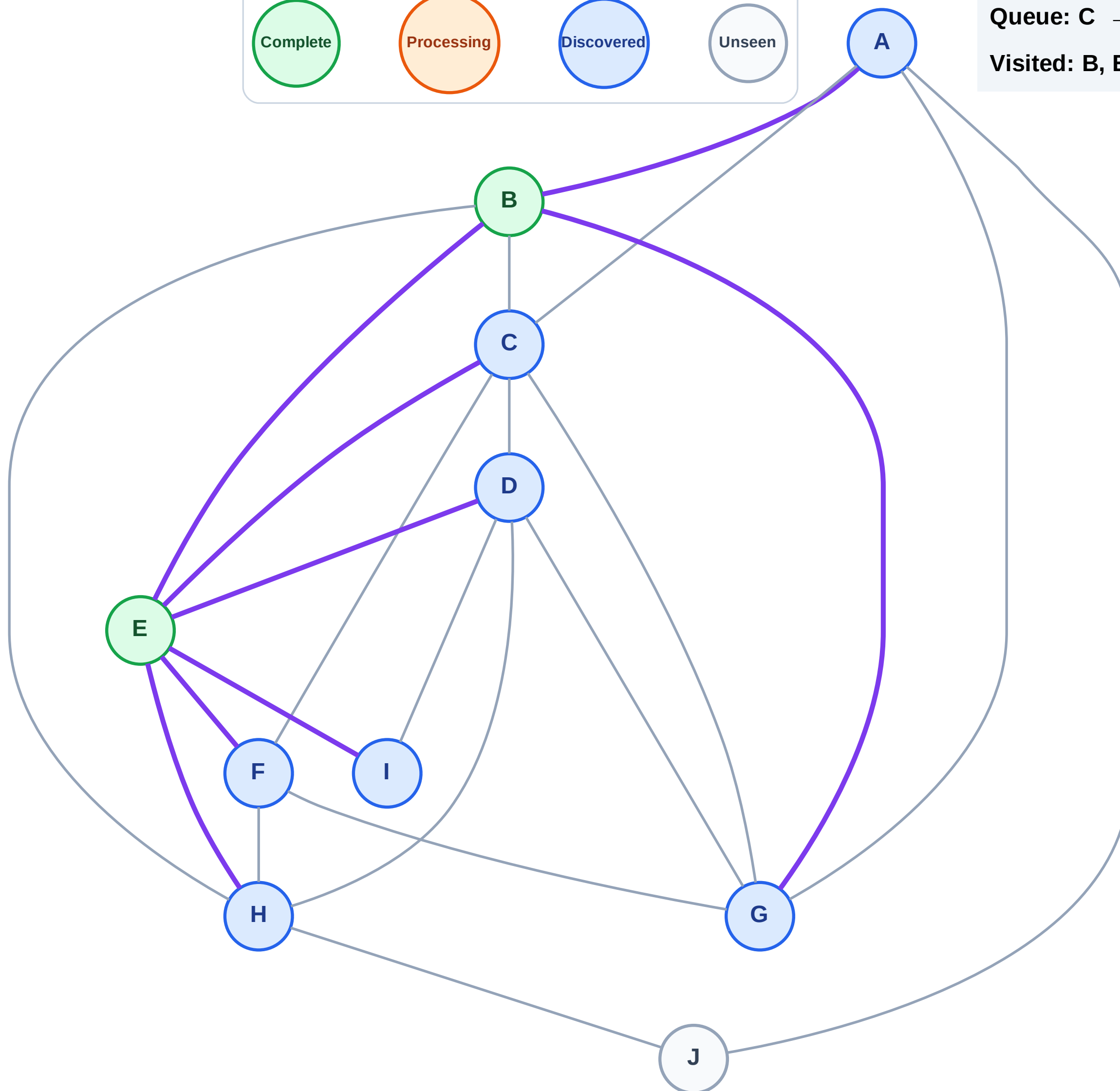
Discovered



Unseen

Queue: C → D → F → H → I → A → G

Visited: B, E



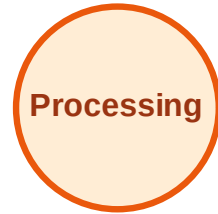
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing

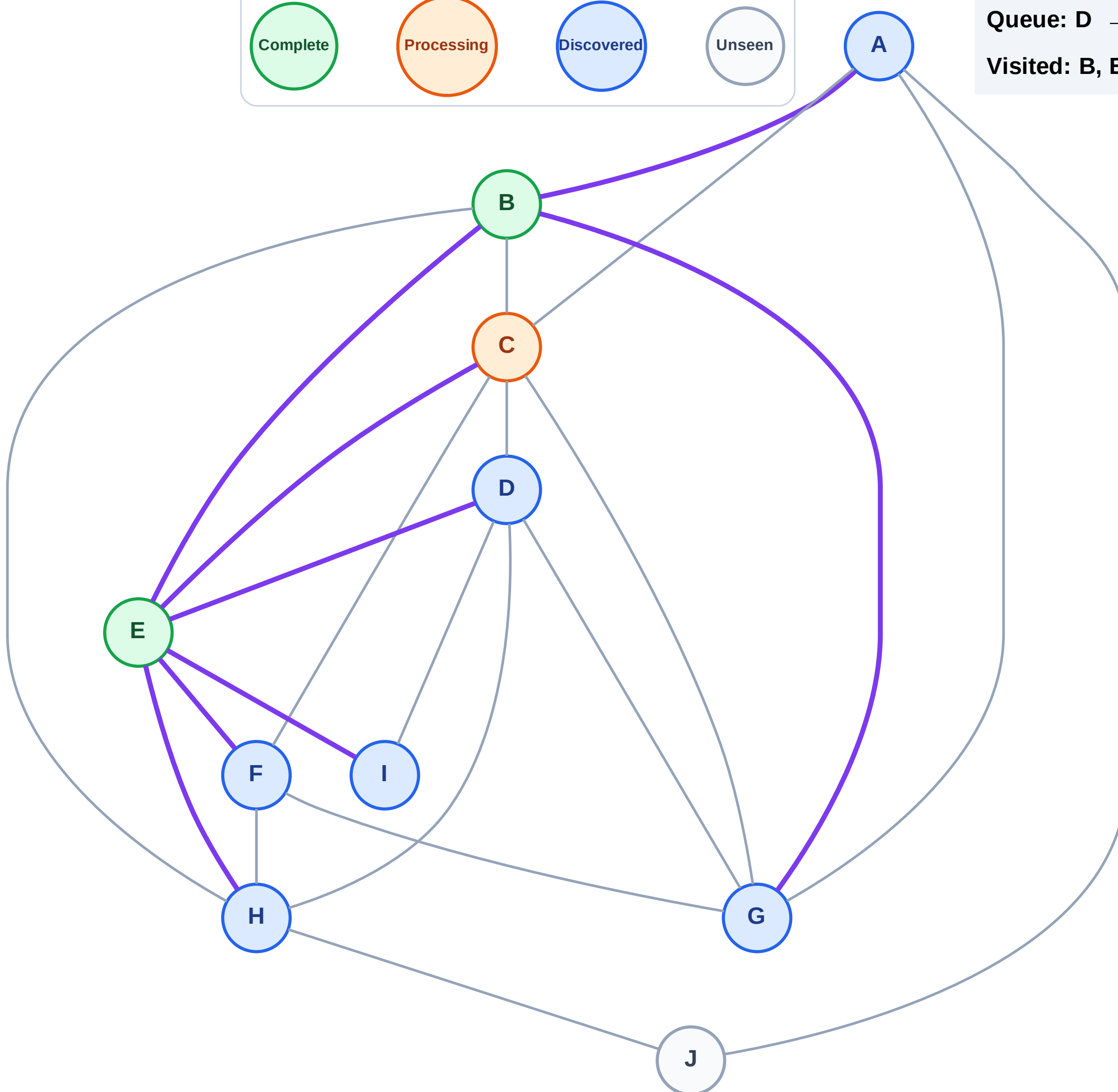


Discovered



Unseen

Queue: D → F → H → I → A → G
Visited: B, E



BFS Traversal

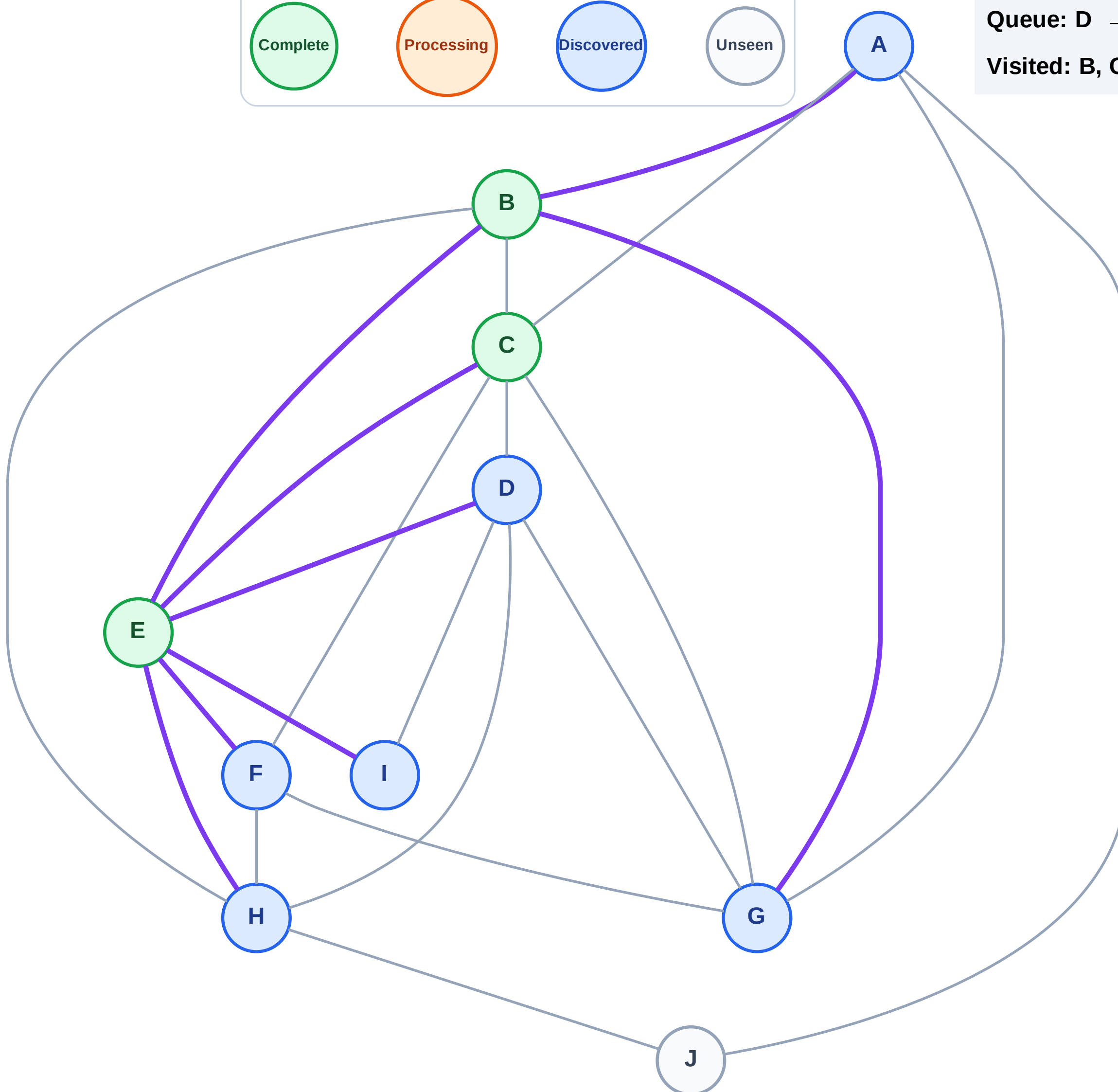
Step 7: No new neighbors from C

Legend



Queue: D → F → H → I → A → G

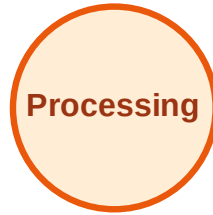
Visited: B, C, E



BFS Traversal

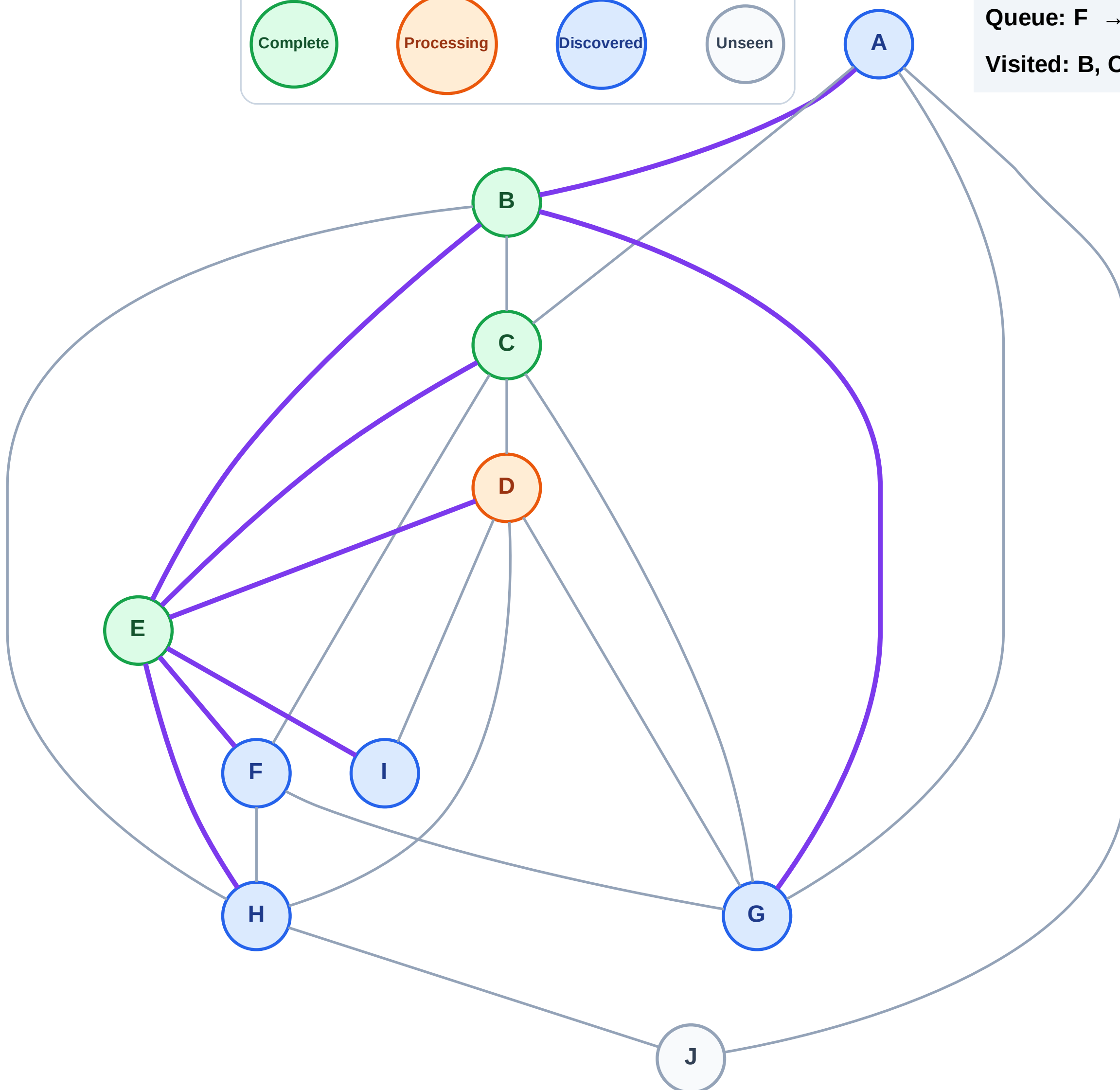
Step 8: Process node D

Legend



Queue: F → H → I → A → G

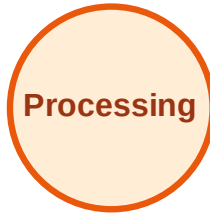
Visited: B, C, E



BFS Traversal

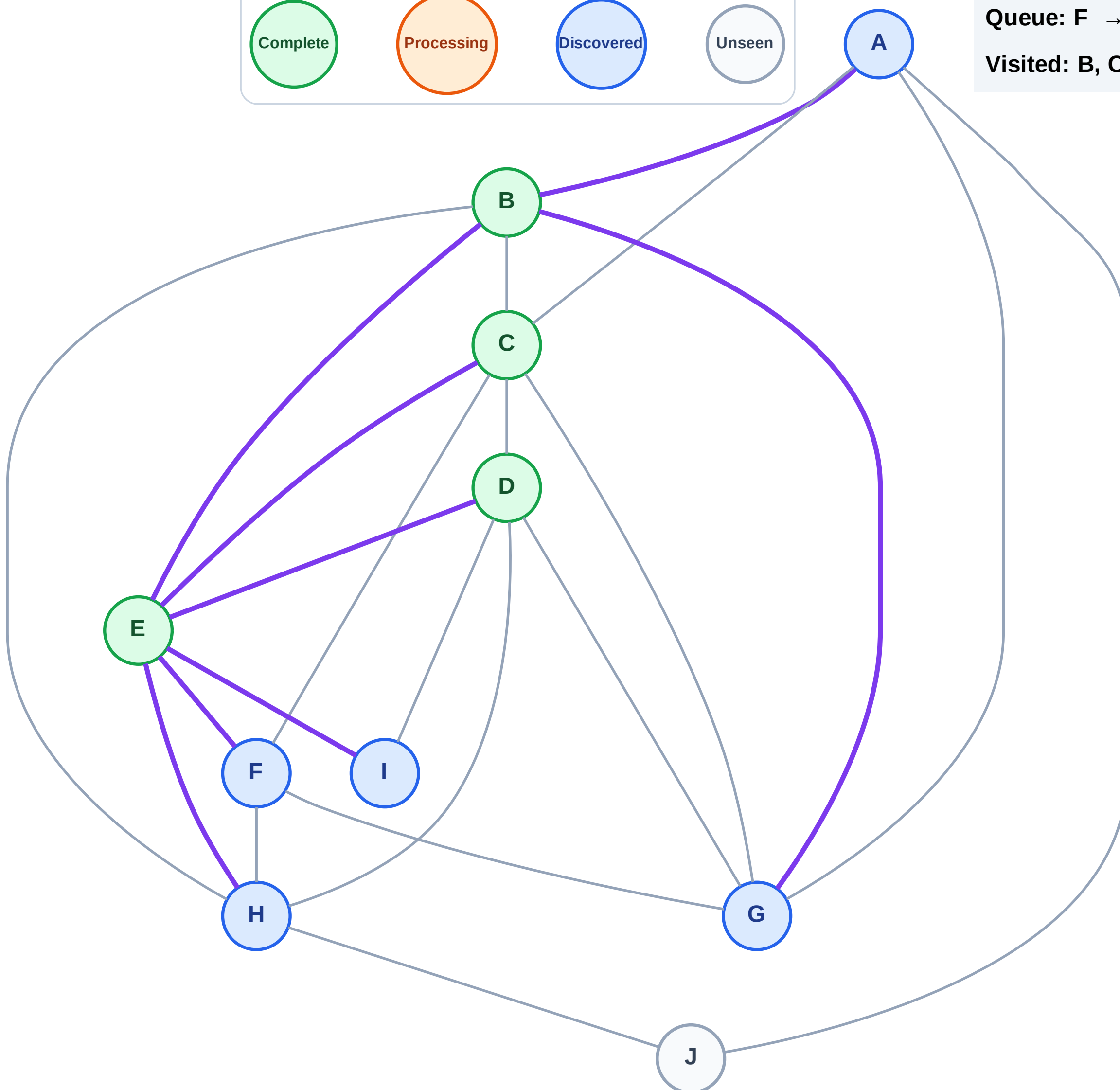
Step 9: No new neighbors from D

Legend



Queue: F → H → I → A → G

Visited: B, C, D, E



BFS Traversal

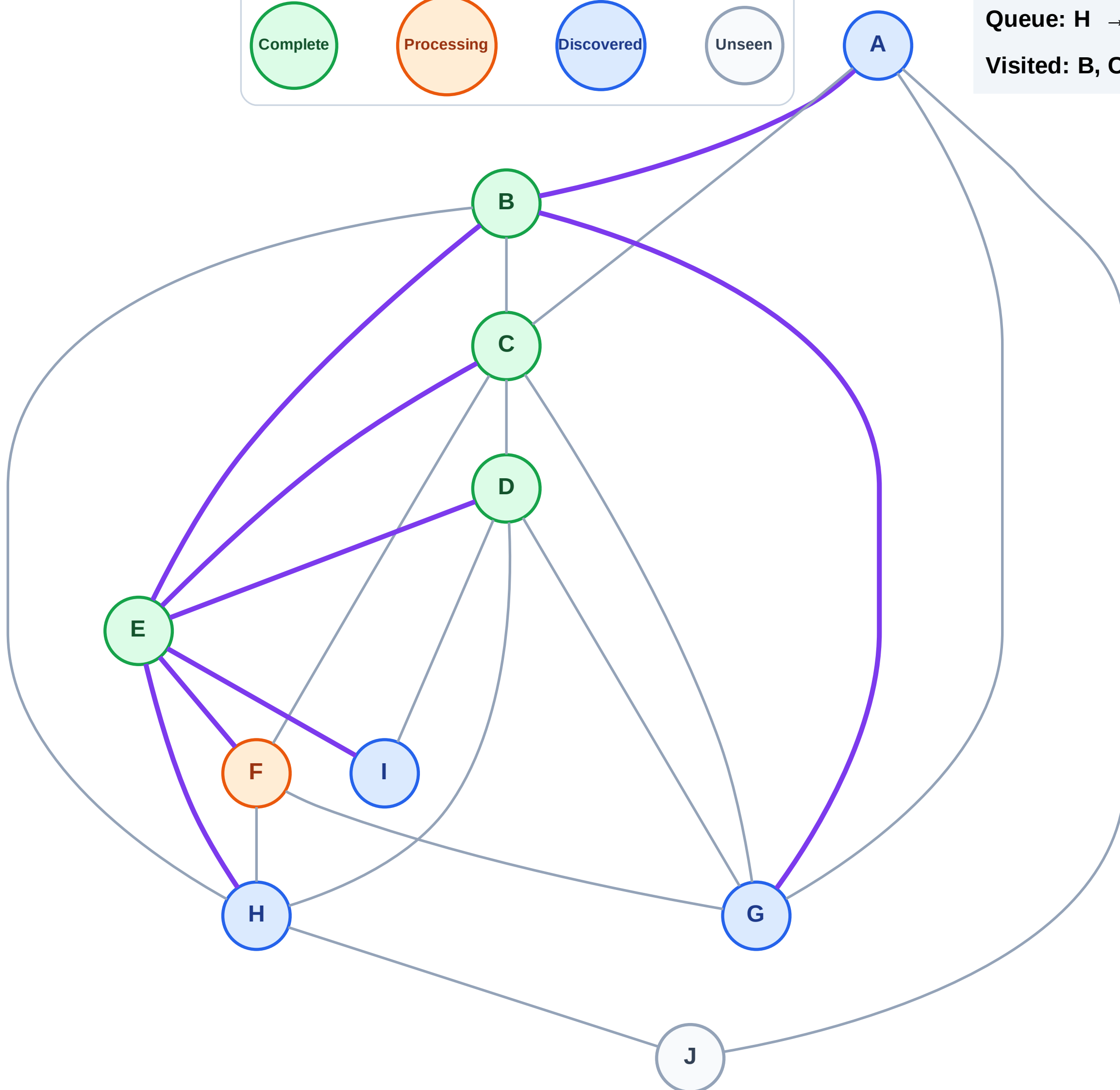
Step 10: Process node F

Legend



Queue: H → I → A → G

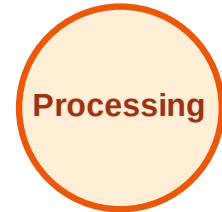
Visited: B, C, D, E



BFS Traversal

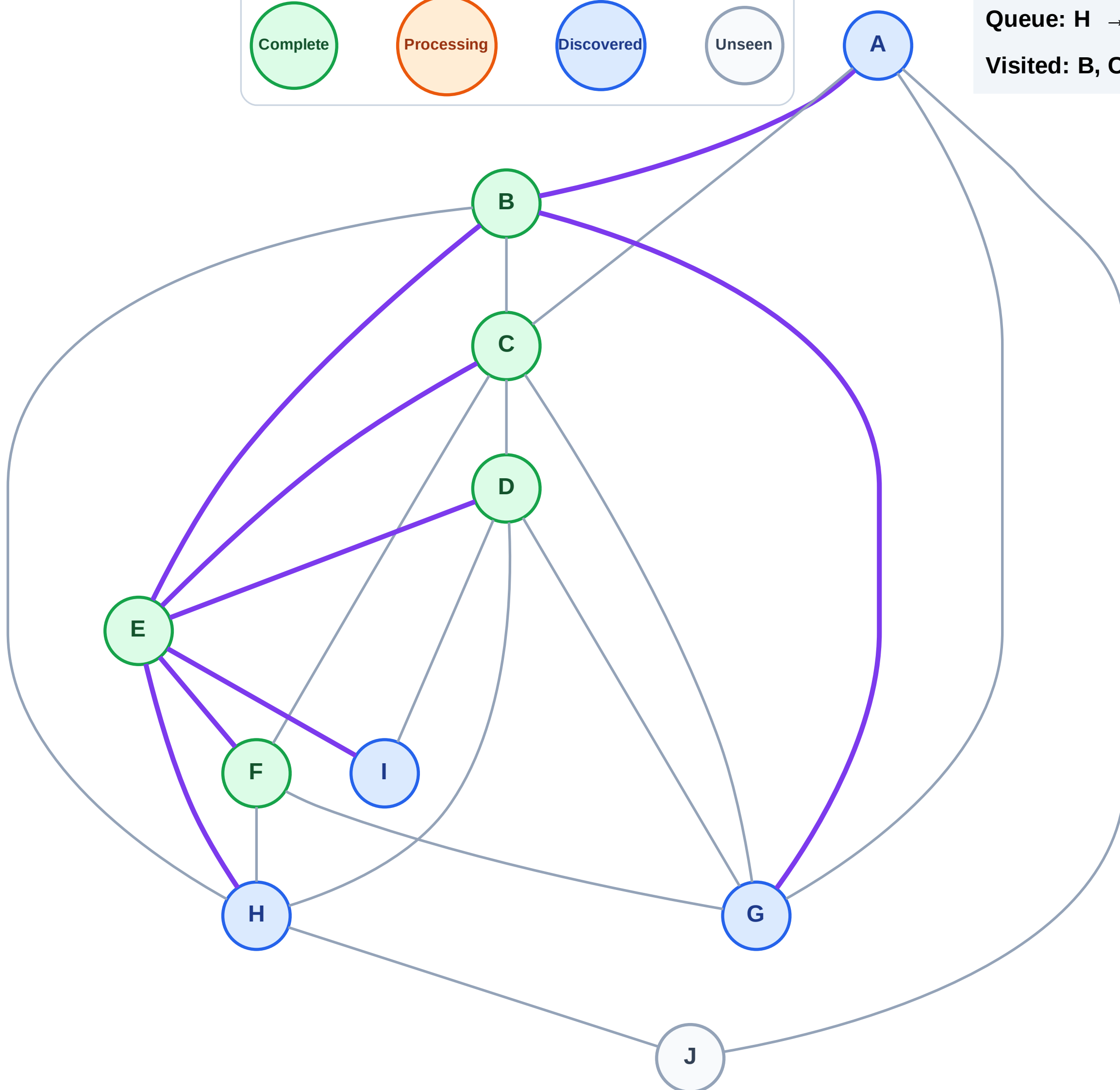
Step 11: No new neighbors from F

Legend



Queue: H → I → A → G

Visited: B, C, D, E, F



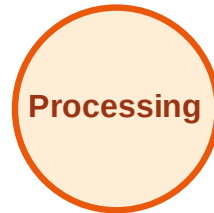
BFS Traversal

Step 12: Process node H

Legend



Complete



Processing



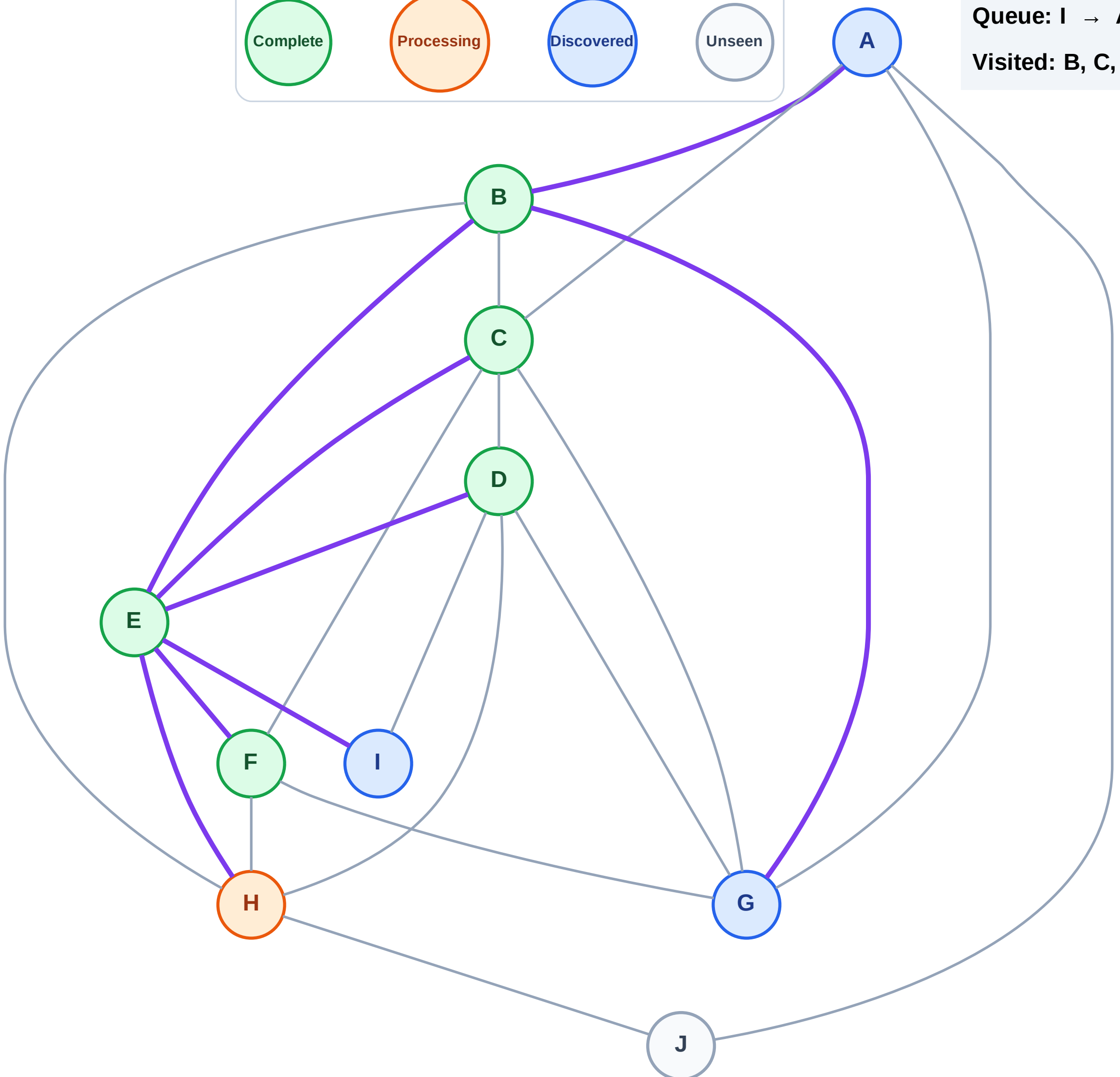
Discovered



Unseen

Queue: I → A → G

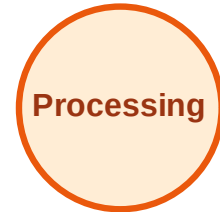
Visited: B, C, D, E, F



BFS Traversal

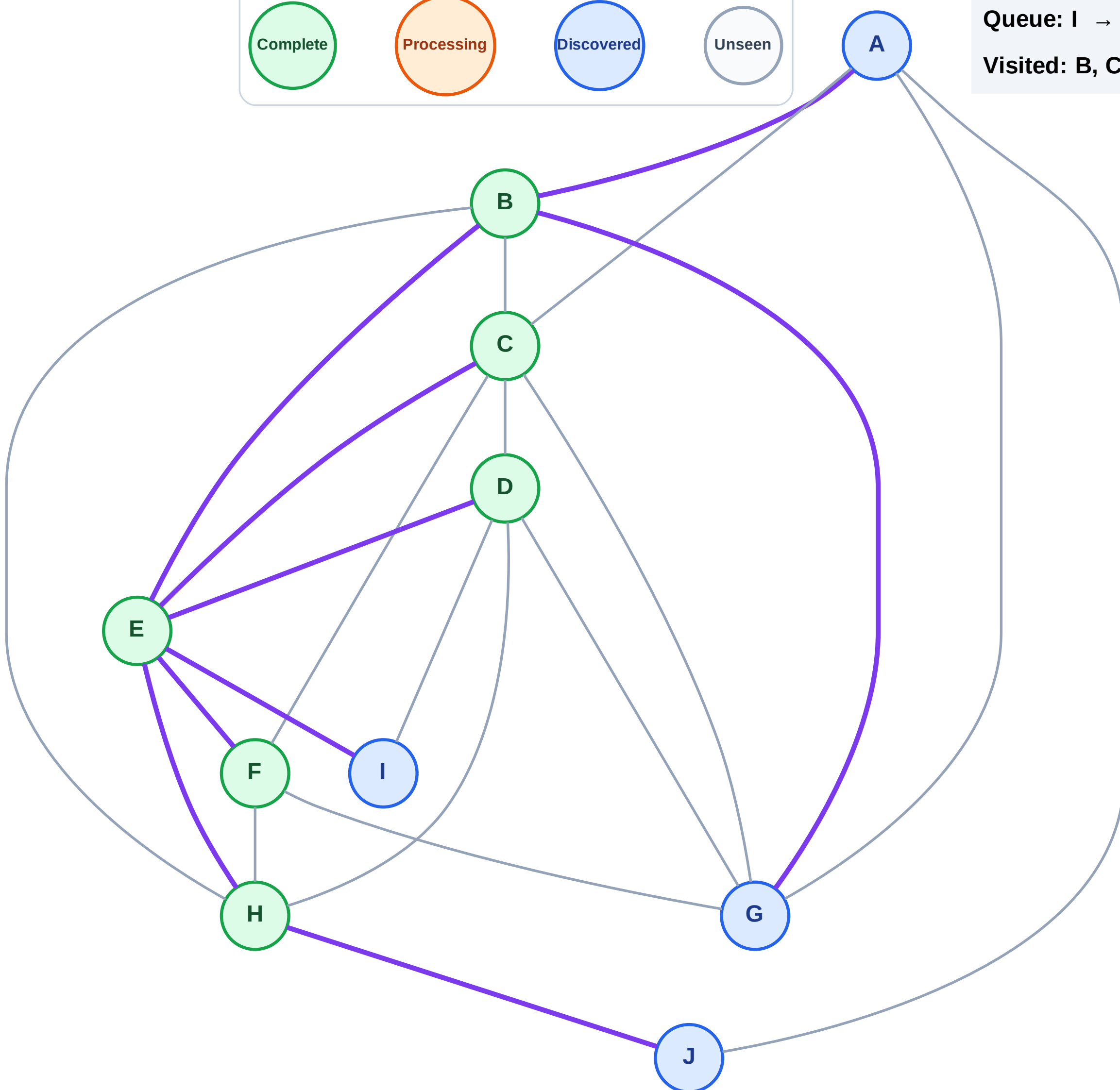
Step 13: Discover J from H

Legend



Queue: I → A → G → J

Visited: B, C, D, E, F, H



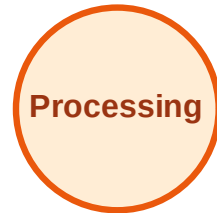
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



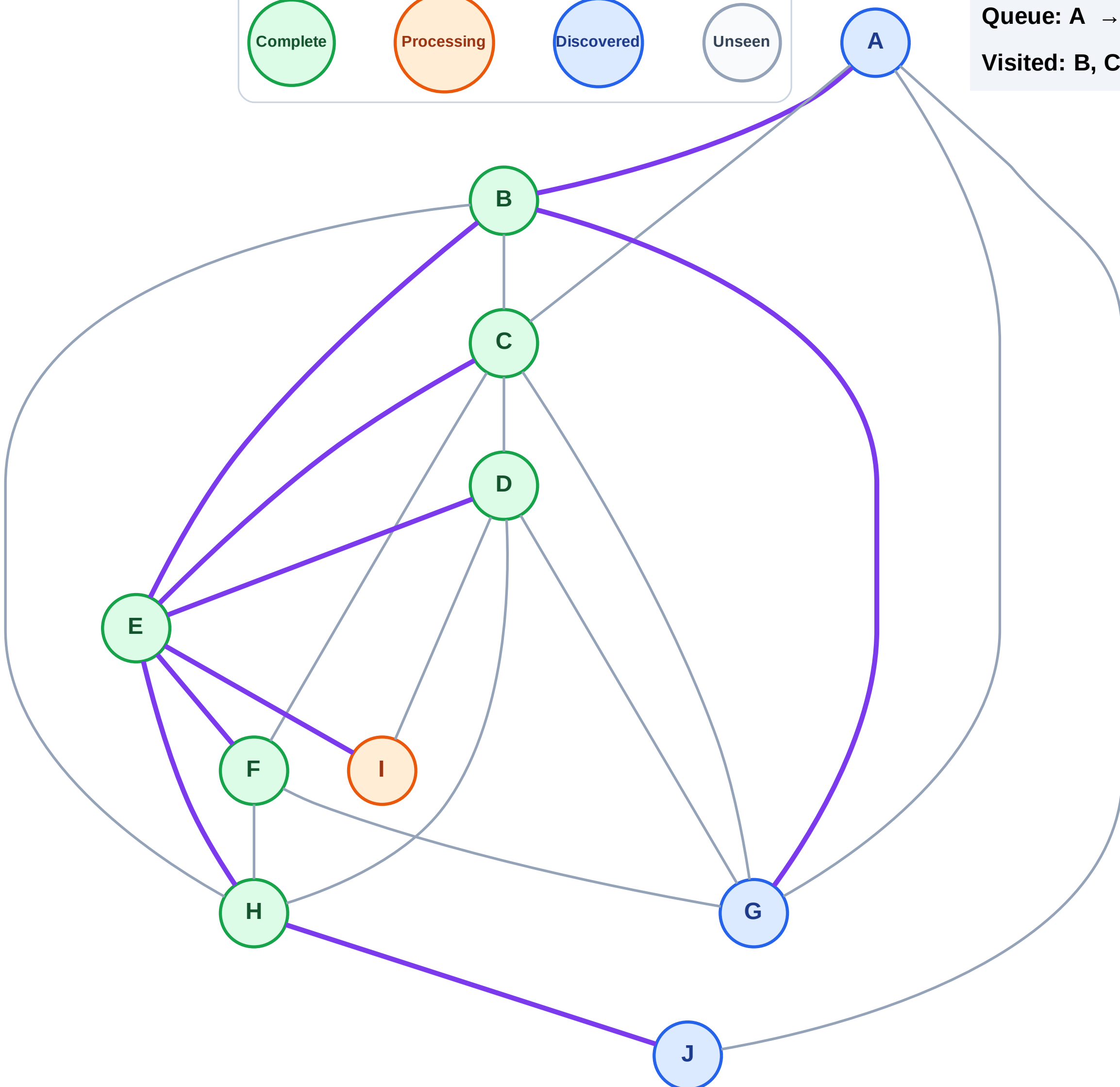
Discovered



Unseen

Queue: A → G → J

Visited: B, C, D, E, F, H



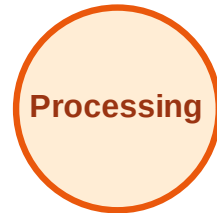
BFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



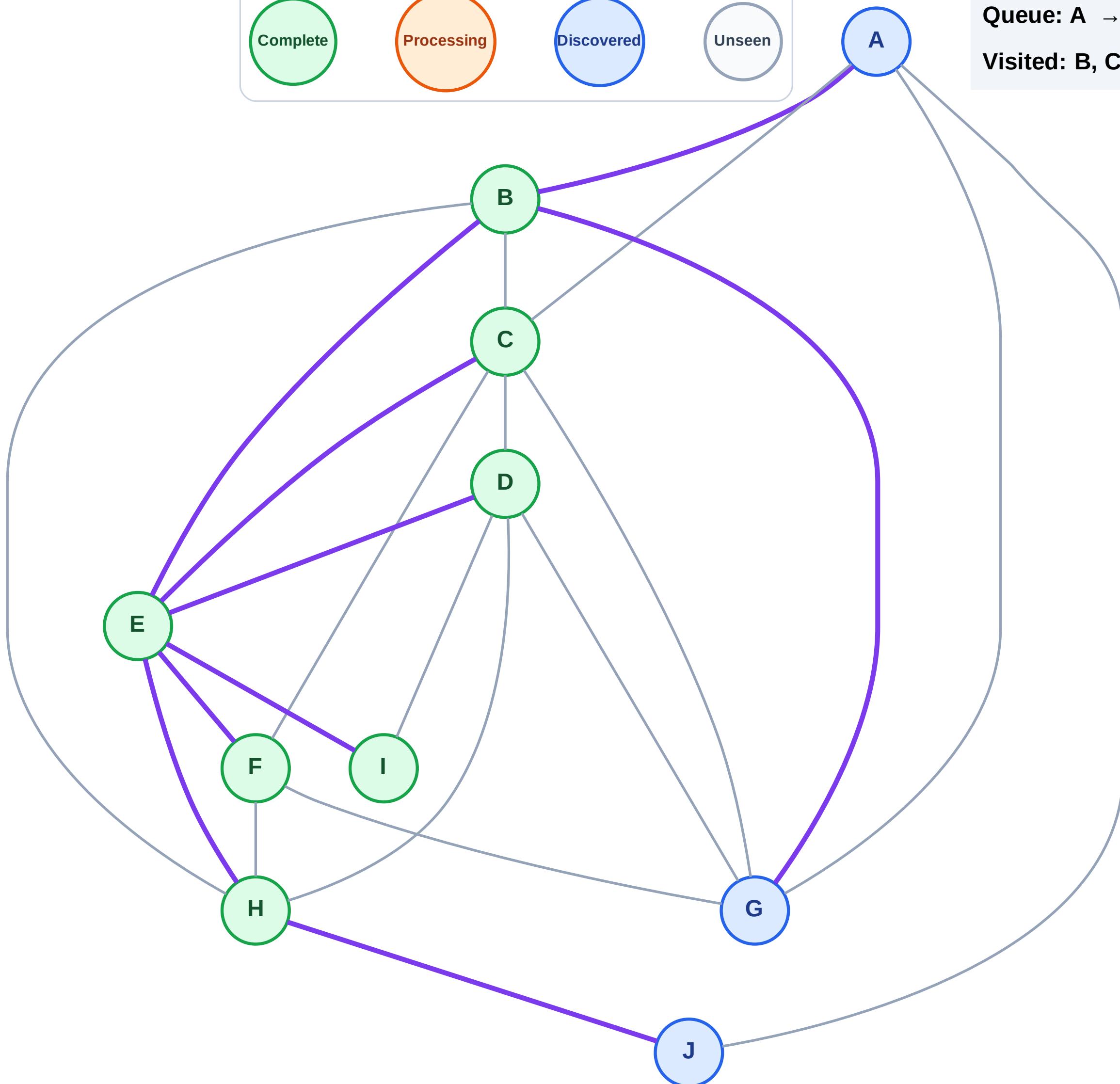
Discovered



Unseen

Queue: A → G → J

Visited: B, C, D, E, F, H, I



BFS Traversal

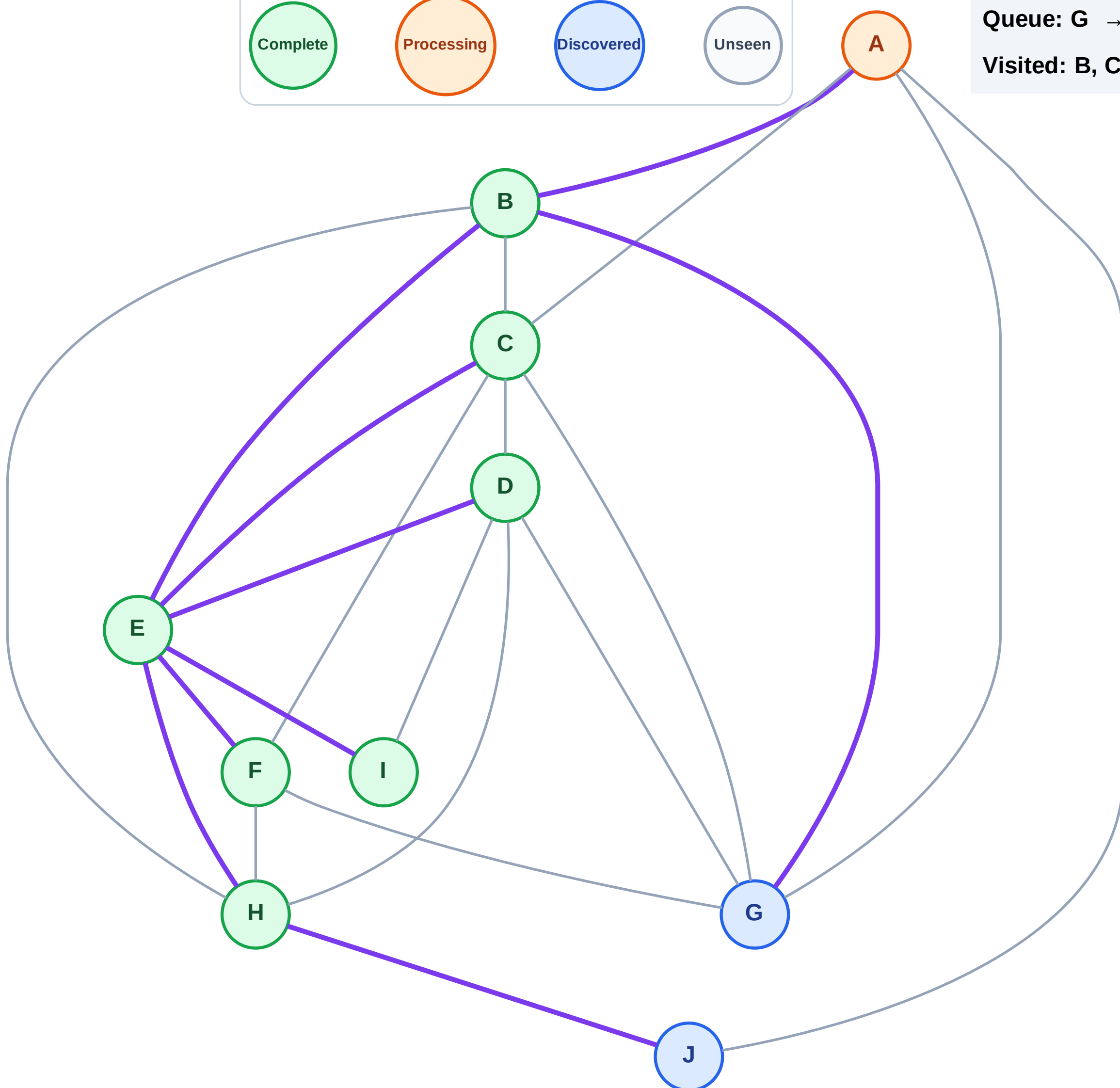
Step 16: Process node A

Legend



Queue: G → J

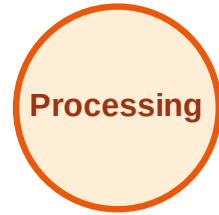
Visited: B, C, D, E, F, H, I



BFS Traversal

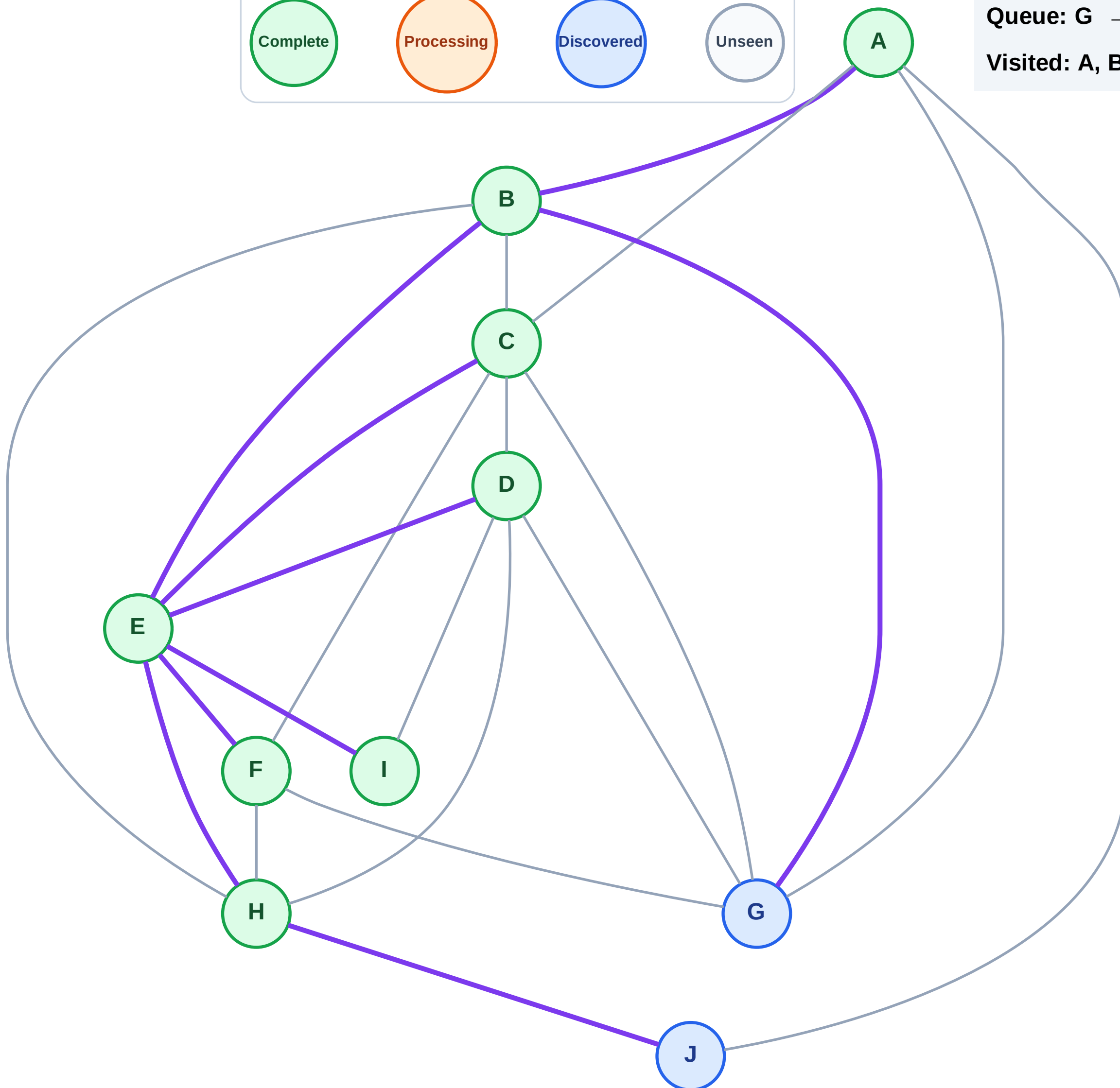
Step 17: No new neighbors from A

Legend



Queue: G → J

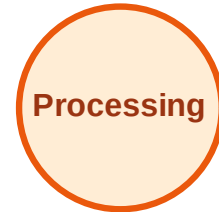
Visited: A, B, C, D, E, F, H, I



BFS Traversal

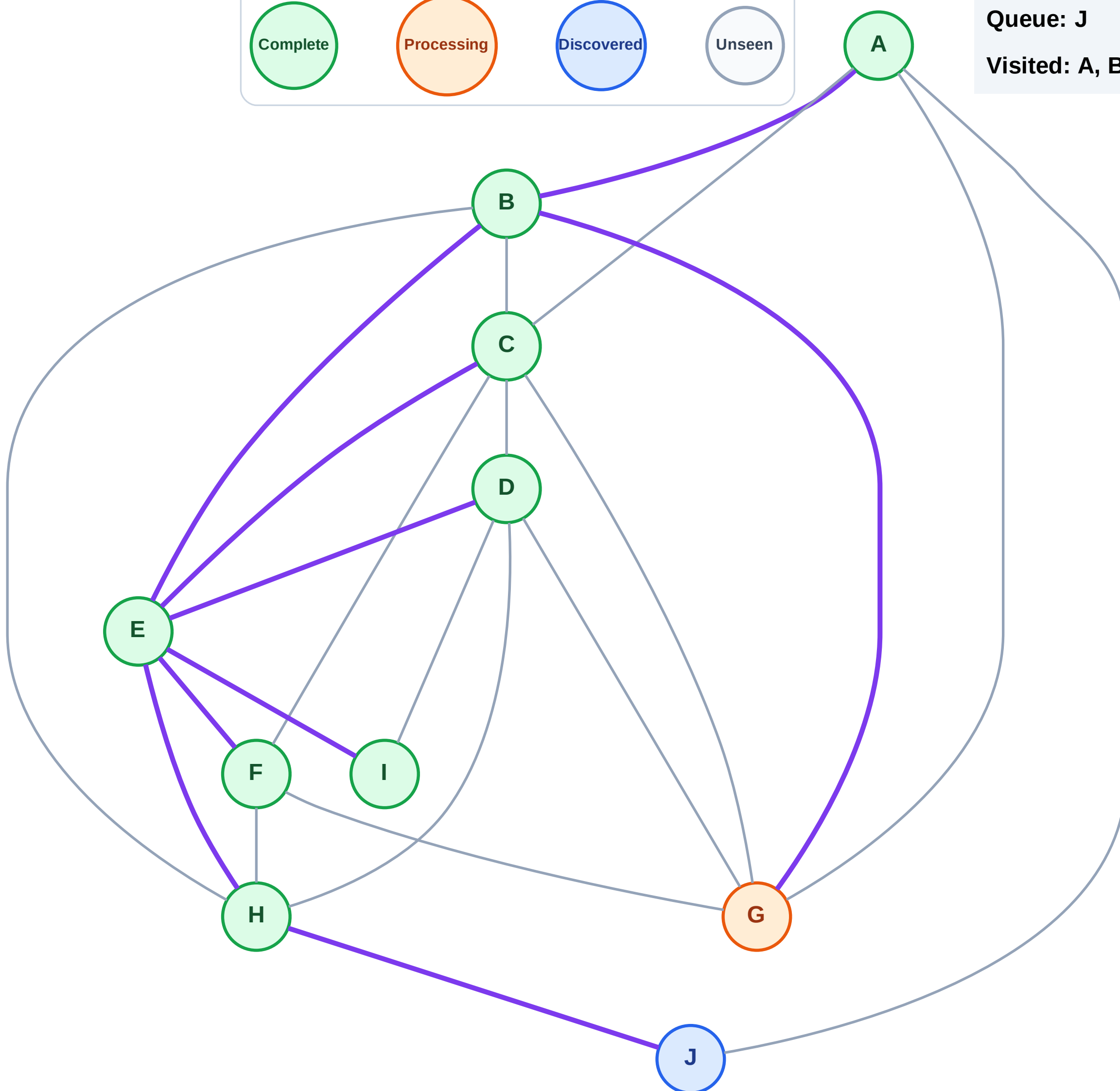
Step 18: Process node G

Legend



Queue: J

Visited: A, B, C, D, E, F, H, I



BFS Traversal

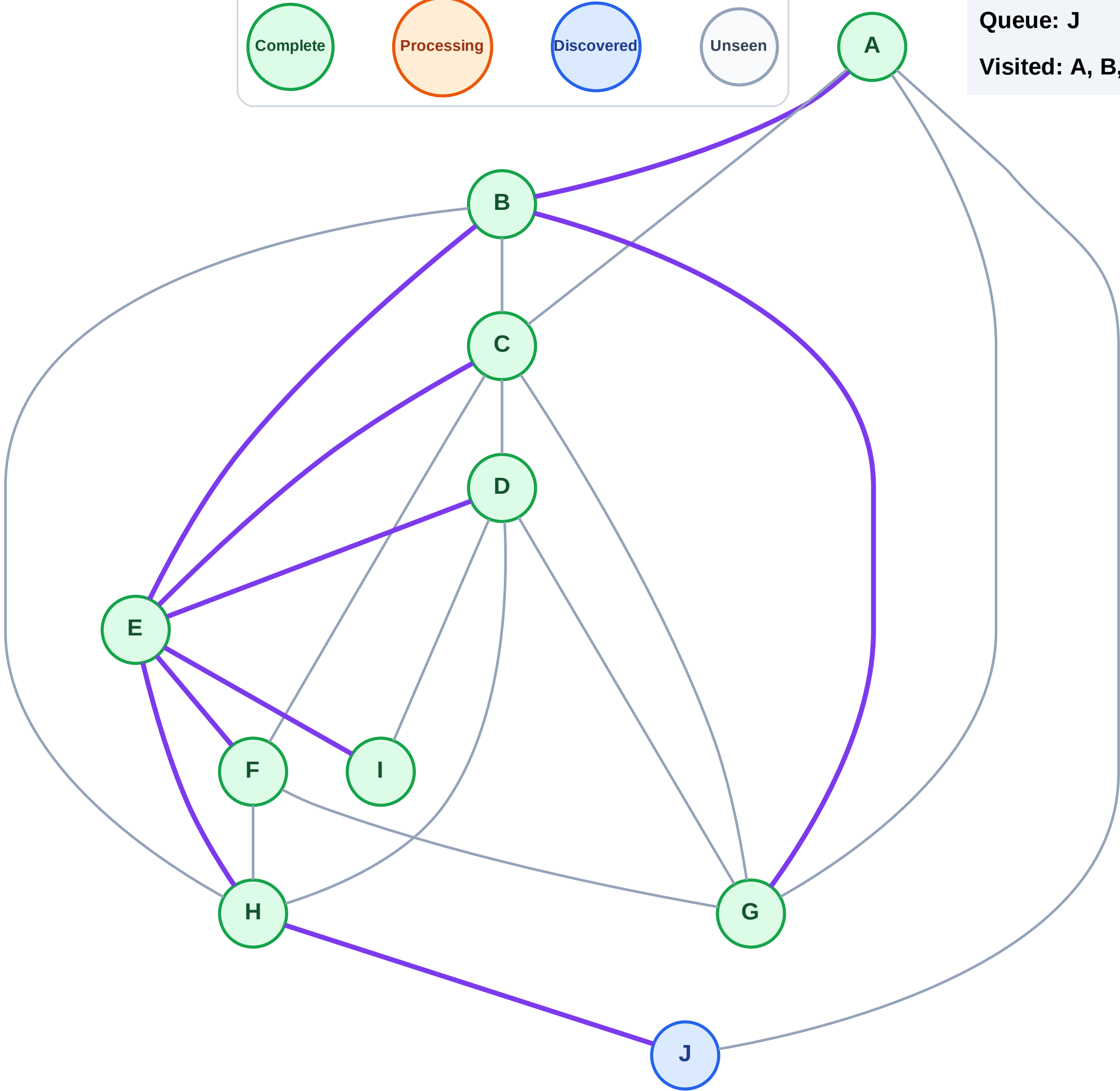
Step 19: No new neighbors from G

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

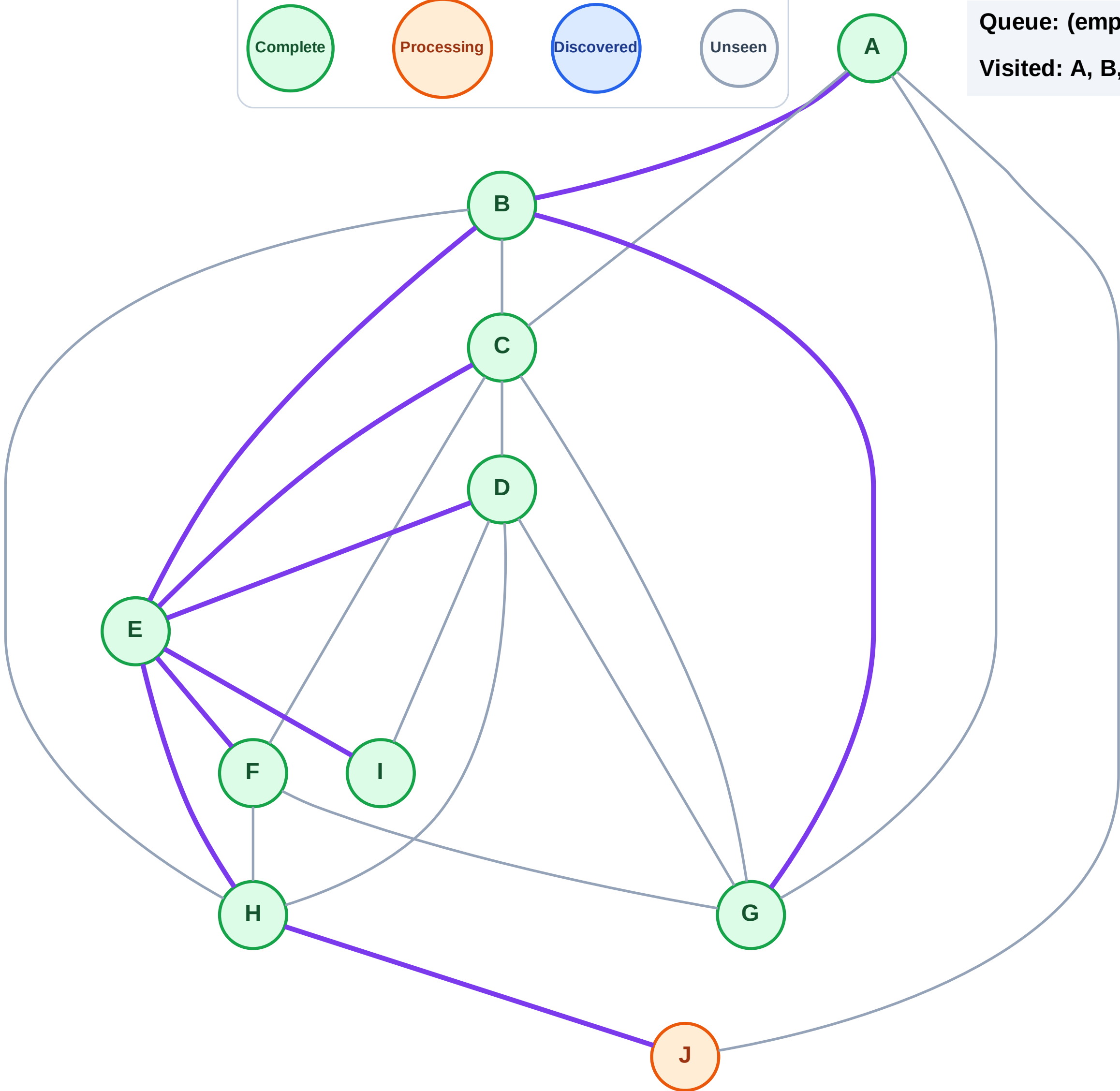
Step 20: Process node J

Legend



Queue: (empty)

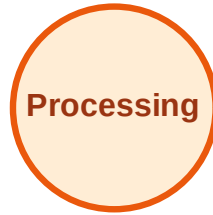
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

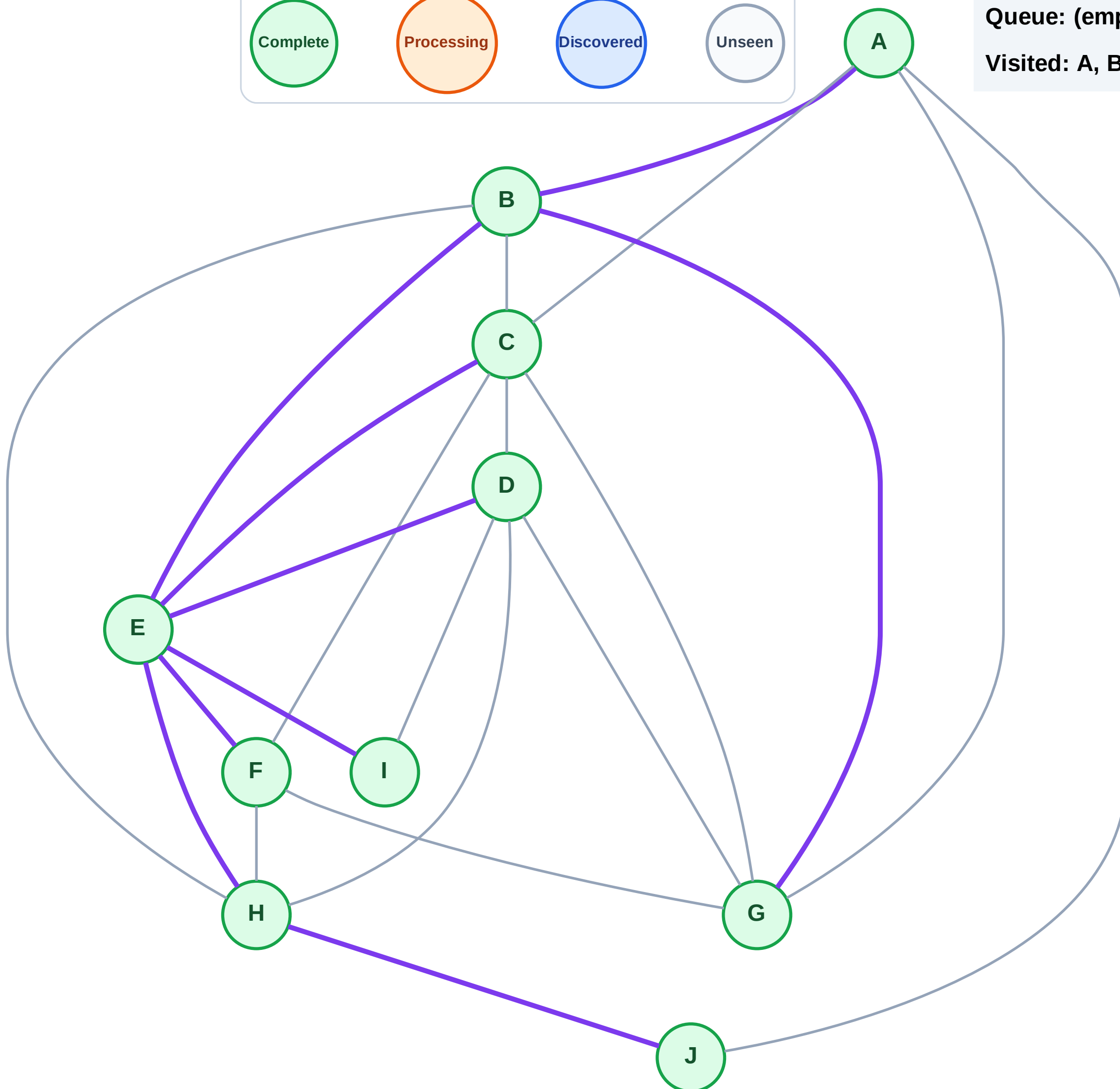
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



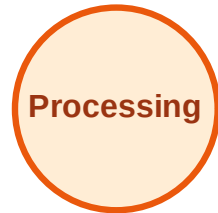
DFS Traversal

Step 1: Start at E

Legend



Complete



Processing



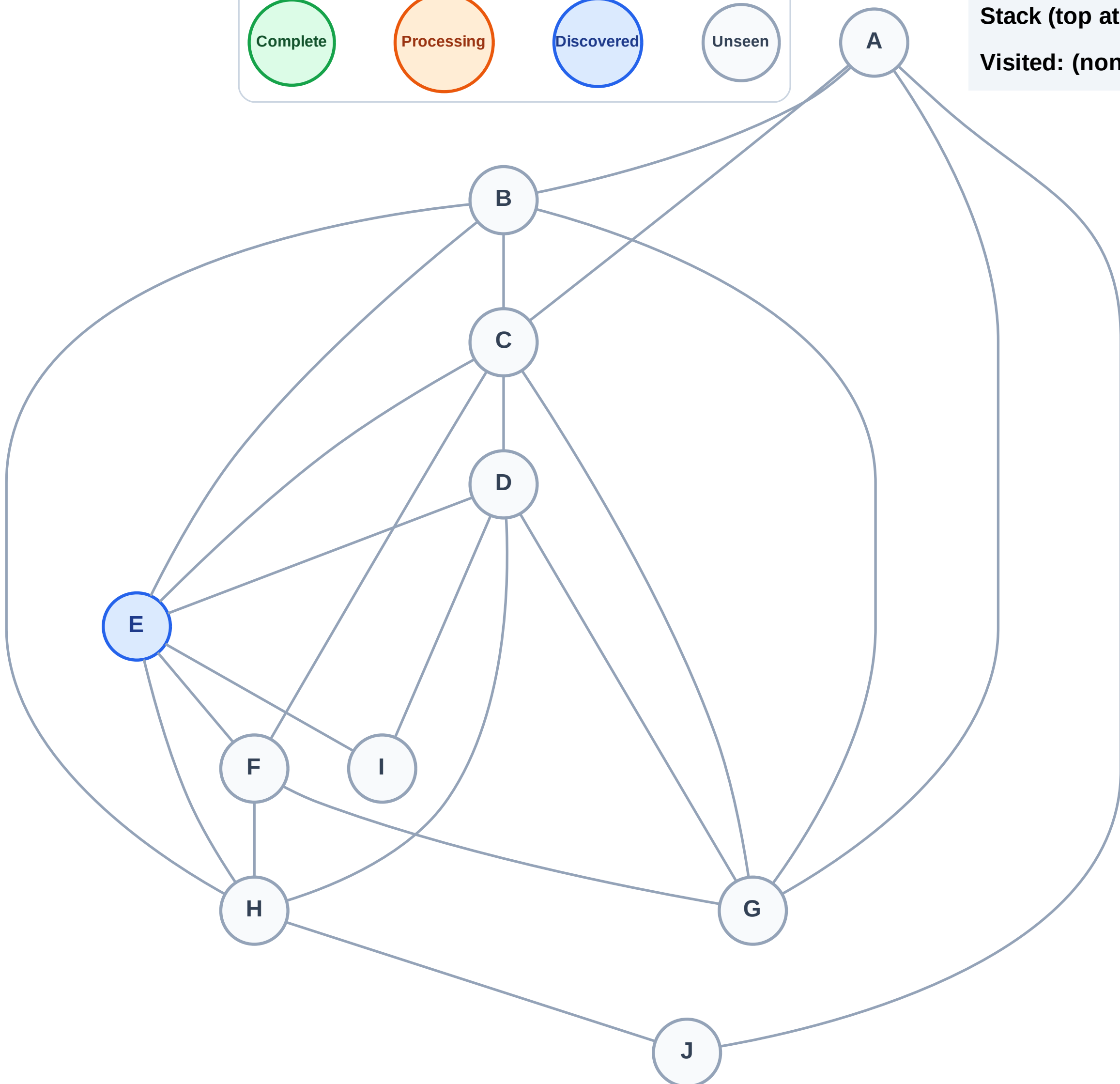
Discovered



Unseen

Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

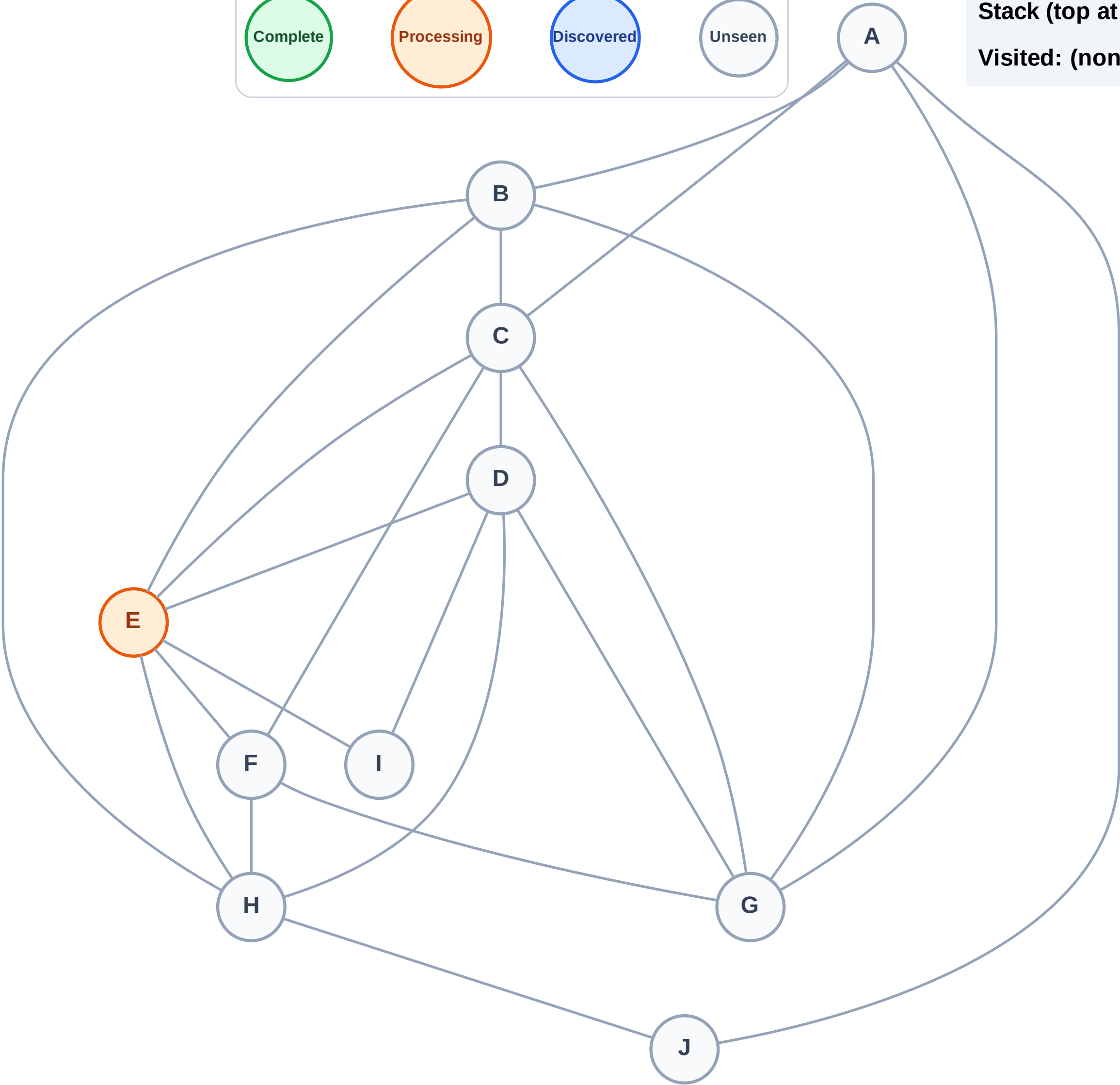
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

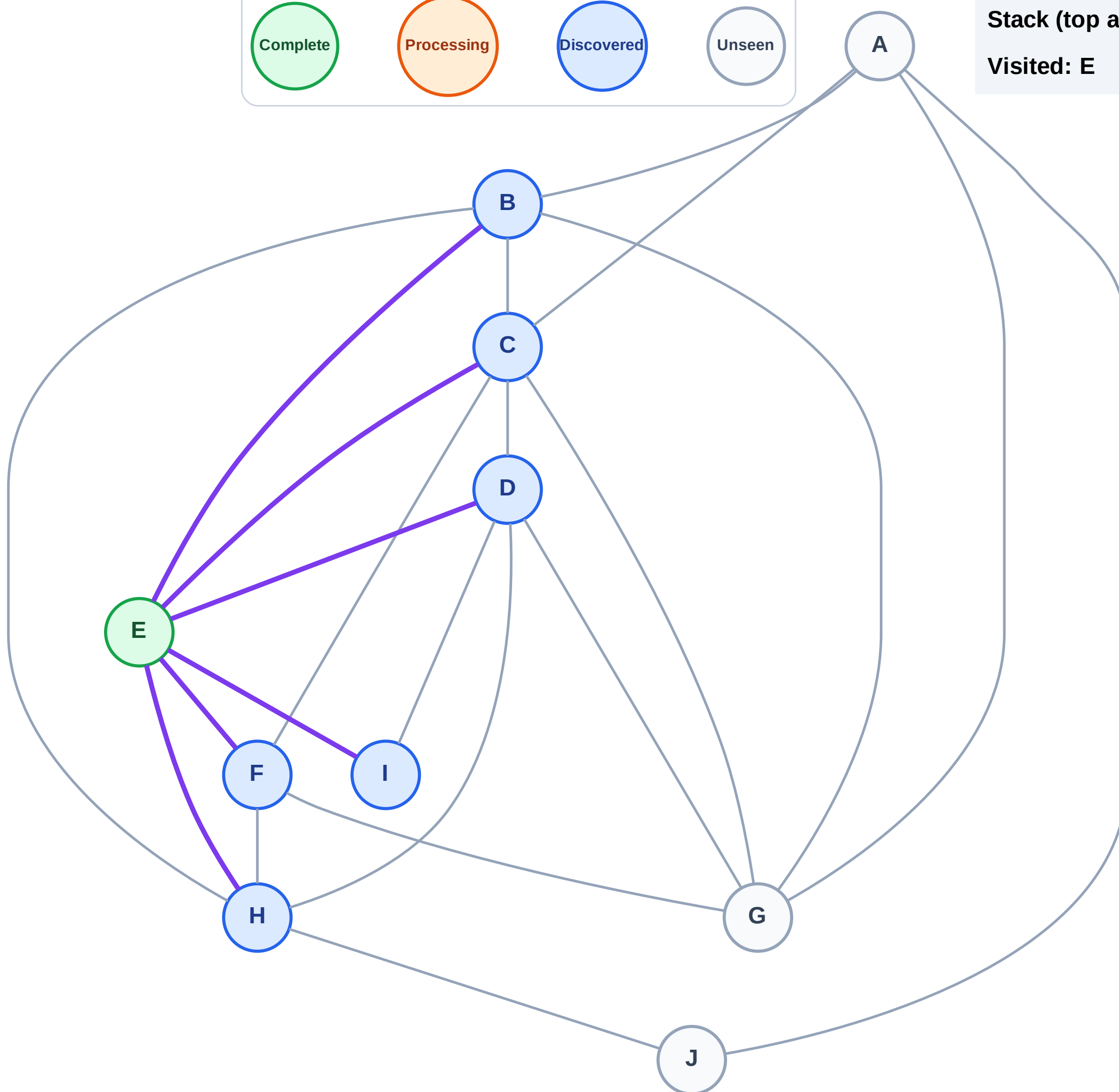
Step 3: Discover I, H, F, D, C, B from E

Legend



Stack (top at right): I | H | F | D | C | B

Visited: E



DFS Traversal

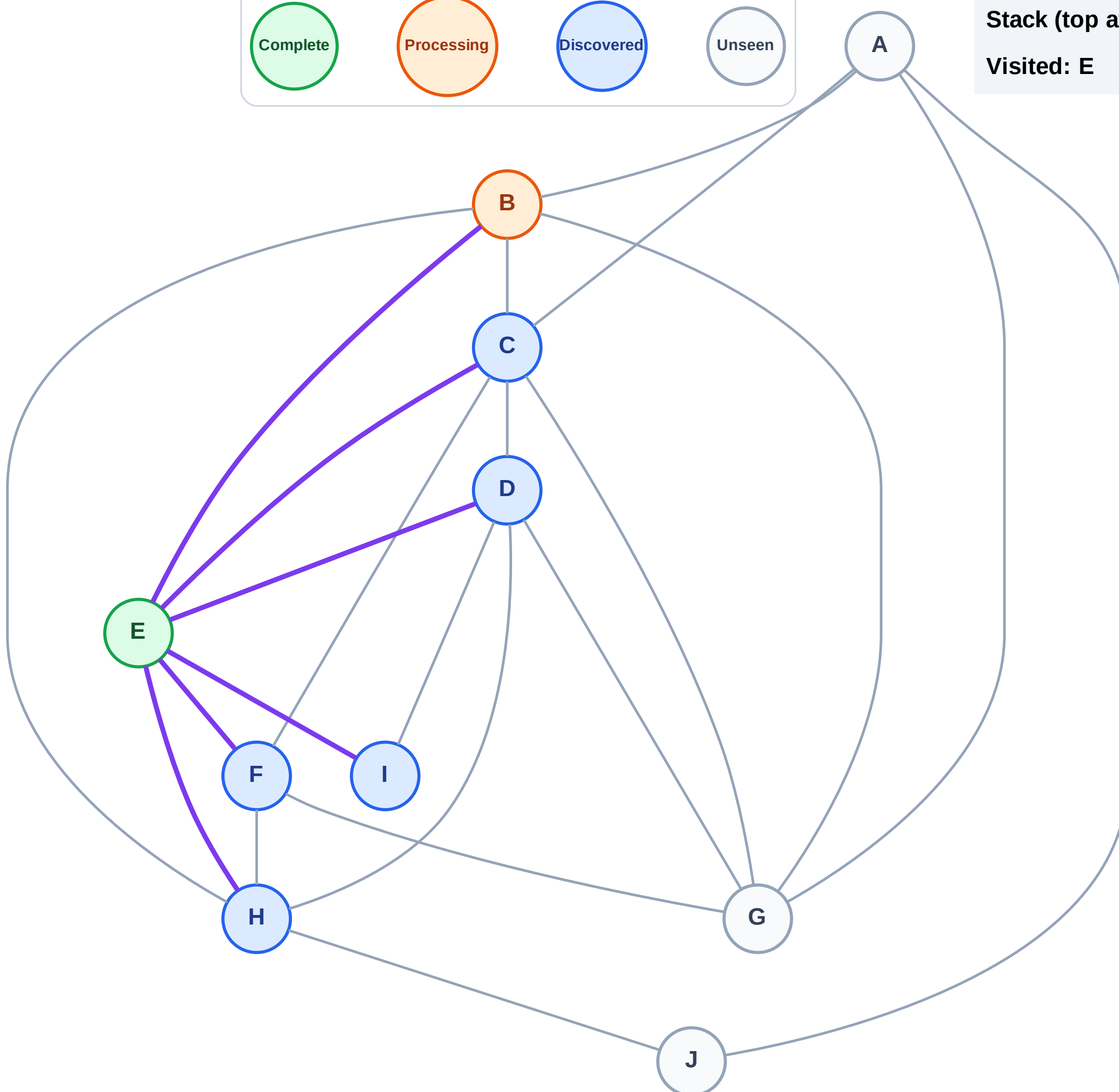
Step 4: Process node B

Legend



Stack (top at right): I | H | F | D | C

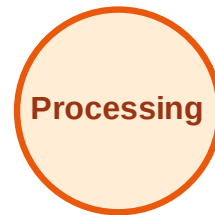
Visited: E



DFS Traversal

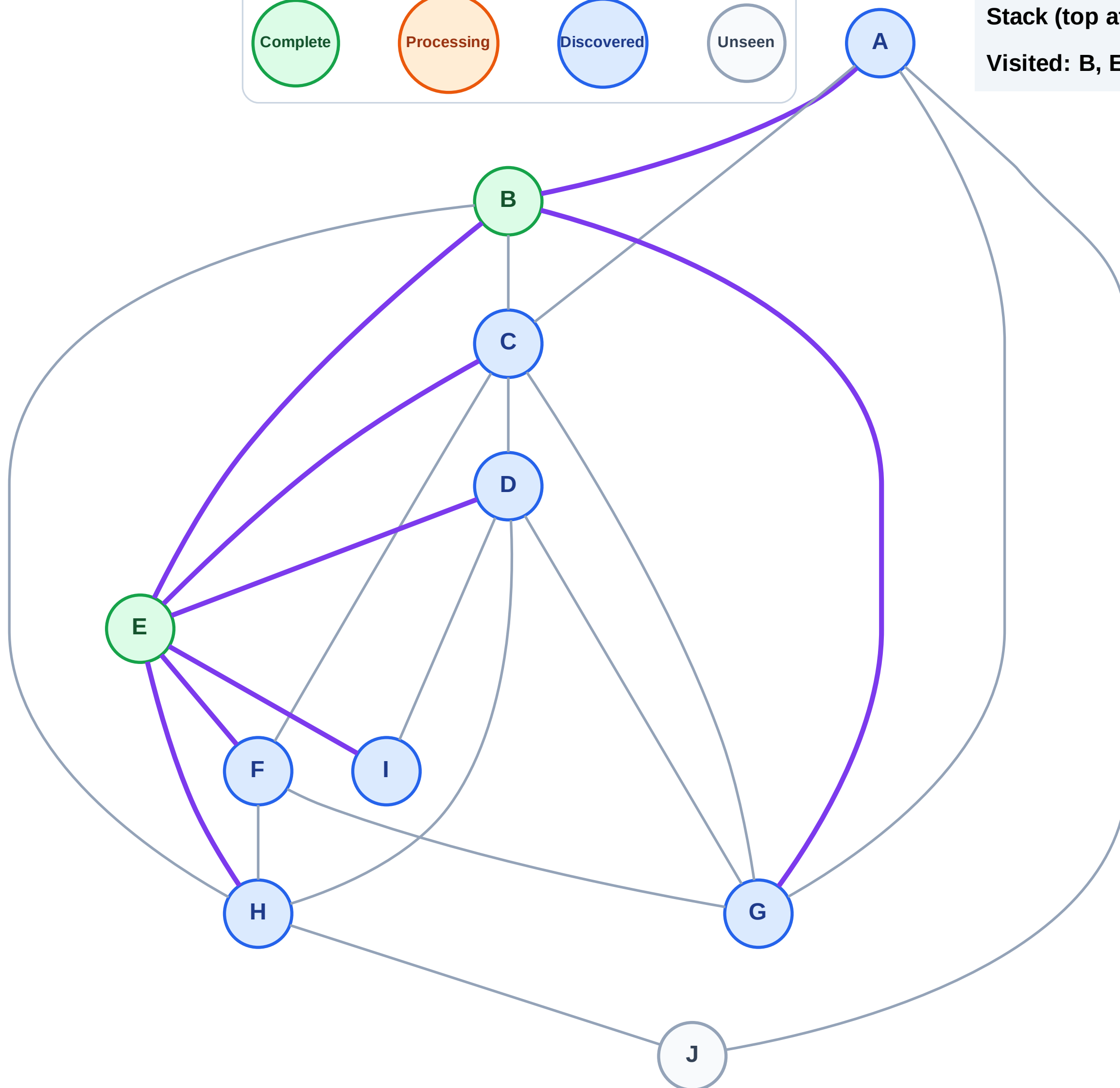
Step 5: Discover G, A from B

Legend



Stack (top at right): I | H | F | D | C | G | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

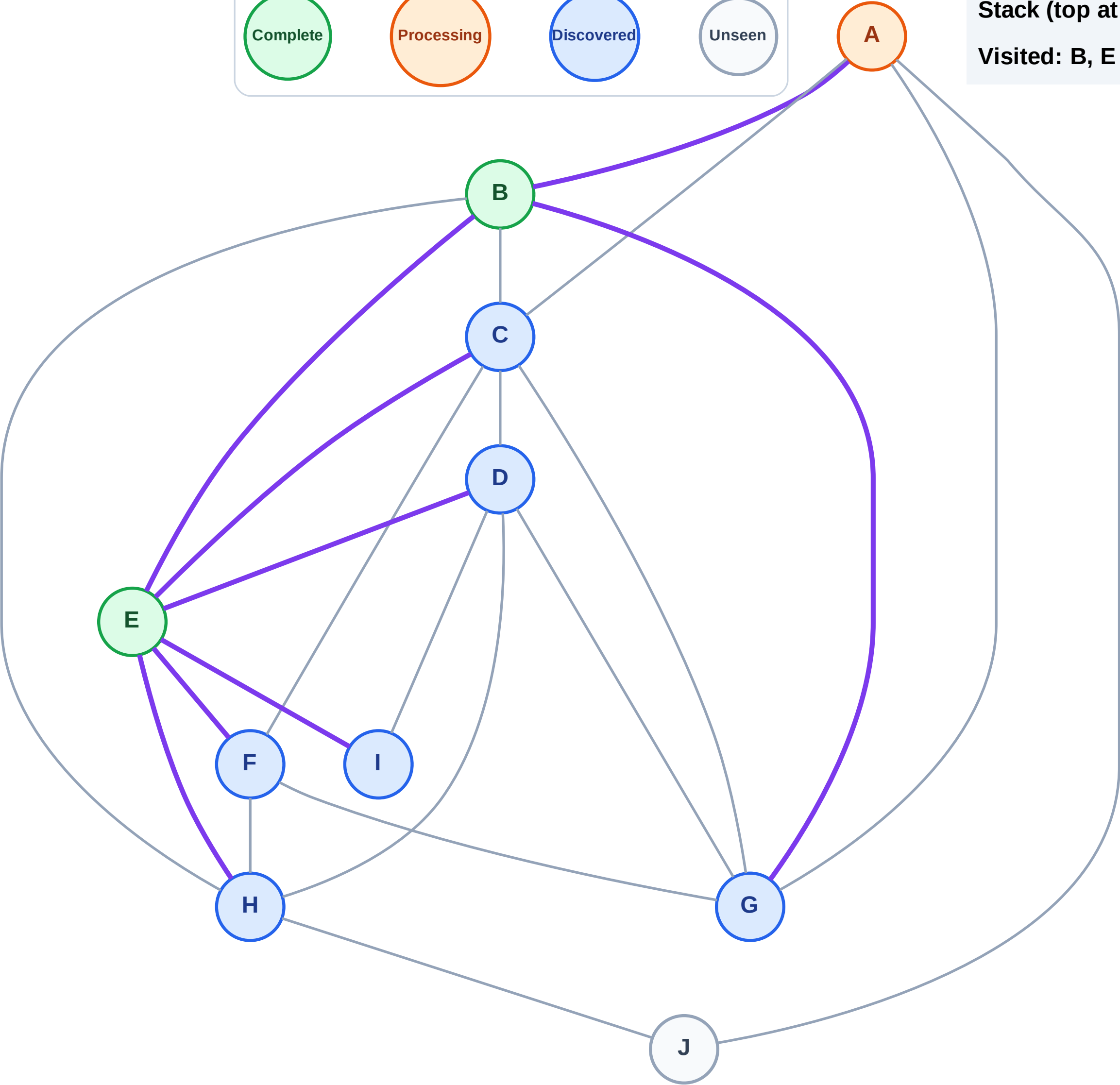
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C | G

Visited: B, E



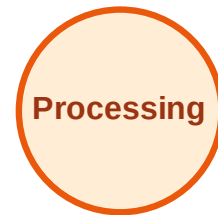
DFS Traversal

Step 7: Discover J from A

Legend



Complete



Processing



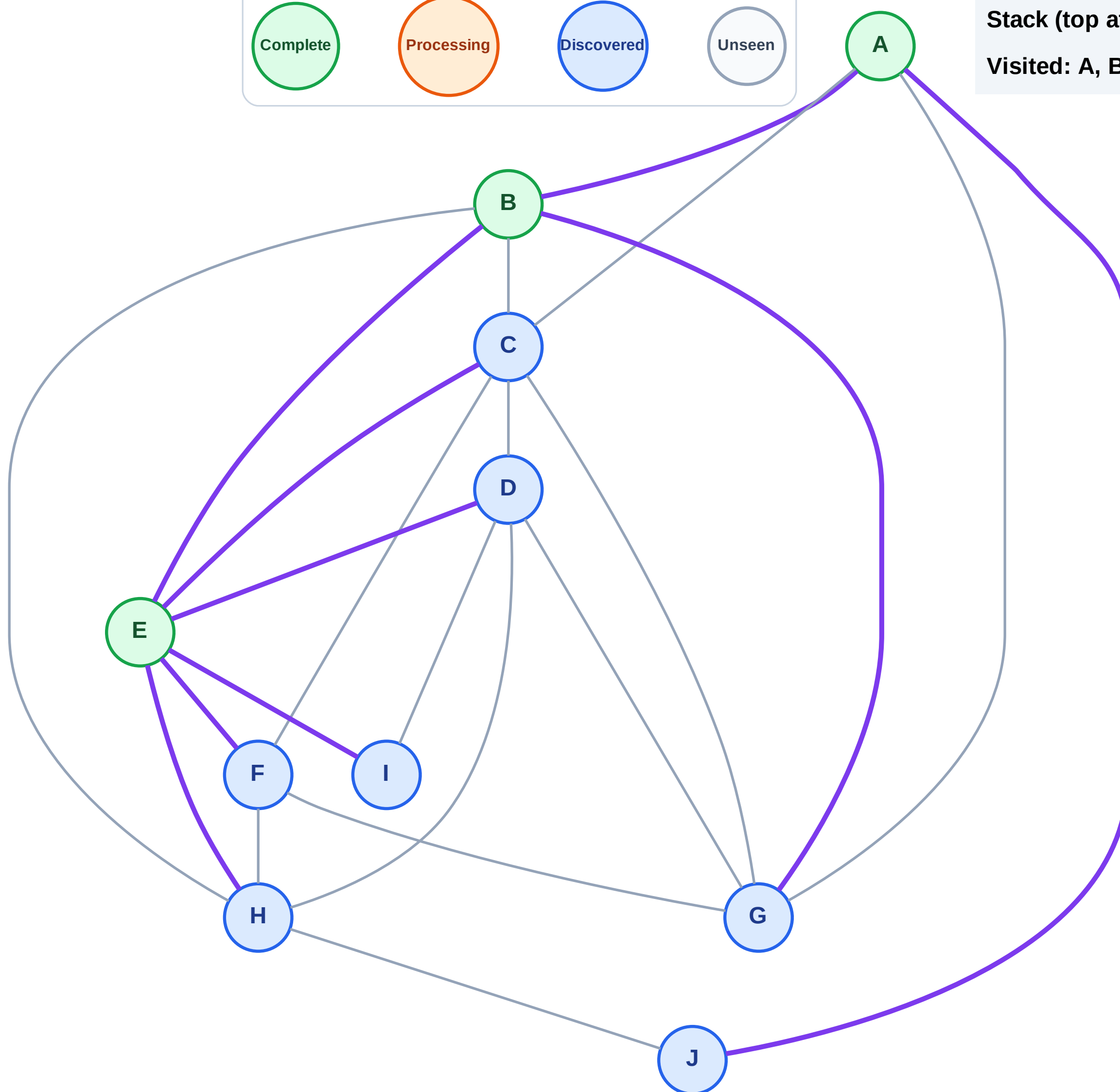
Discovered



Unseen

Stack (top at right): I | H | F | D | C | G | J

Visited: A, B, E



DFS Traversal

Step 8: Process node J

Legend

Complete

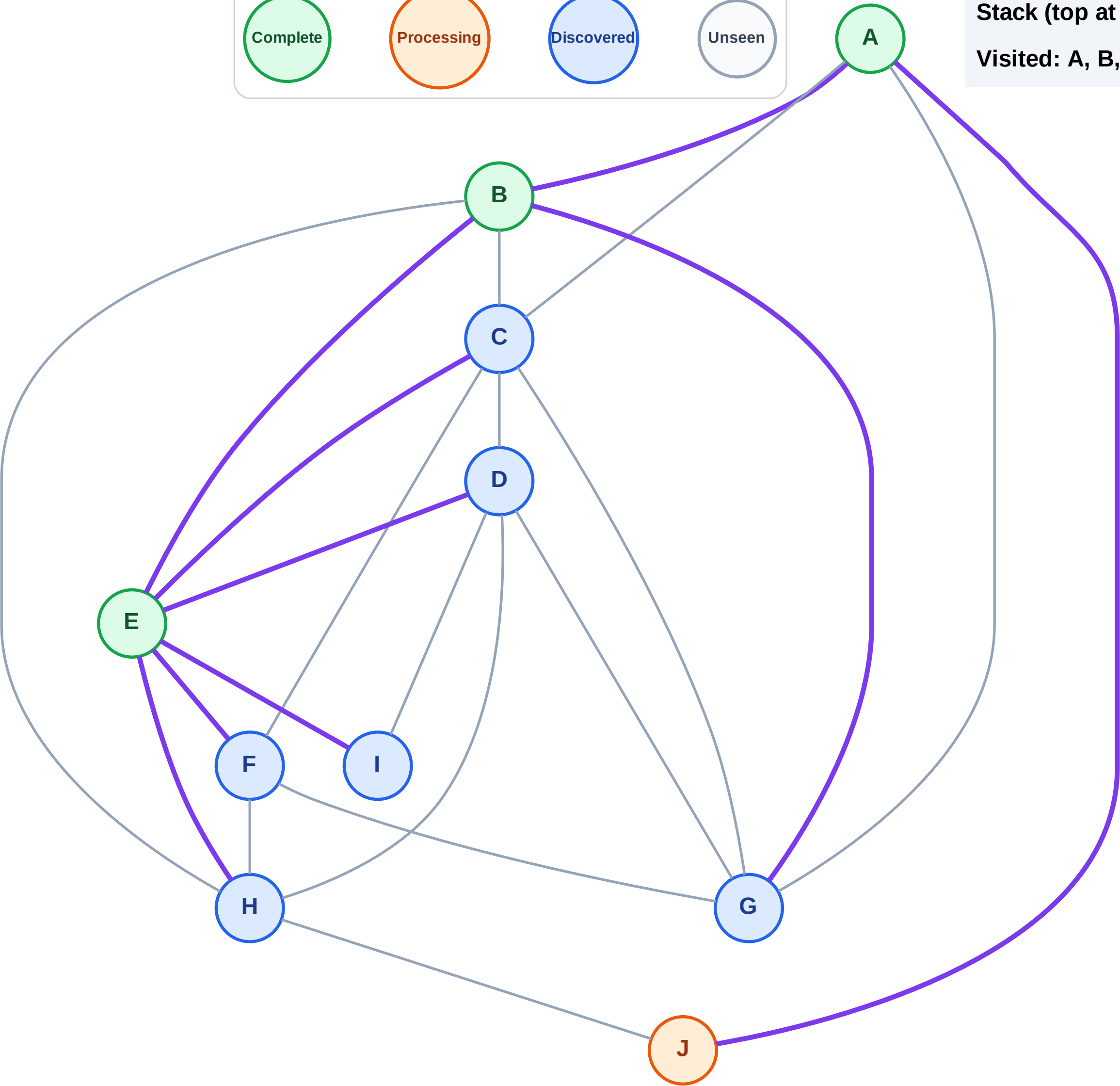
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C | G

Visited: A, B, E



DFS Traversal

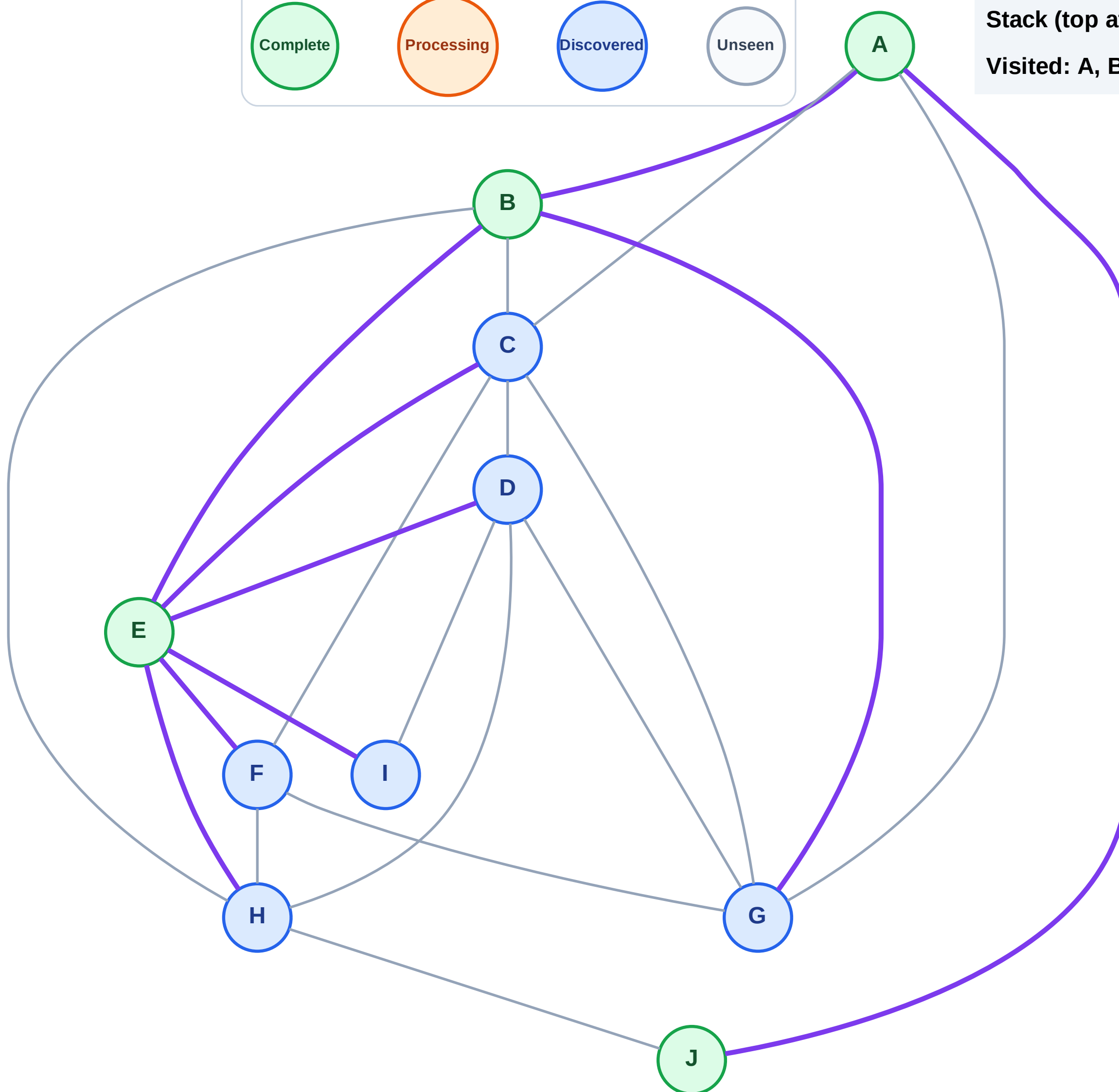
Step 9: No new neighbors from J

Legend



Stack (top at right): I | H | F | D | C | G

Visited: A, B, E, J



DFS Traversal

Step 10: Process node G

Legend

Complete

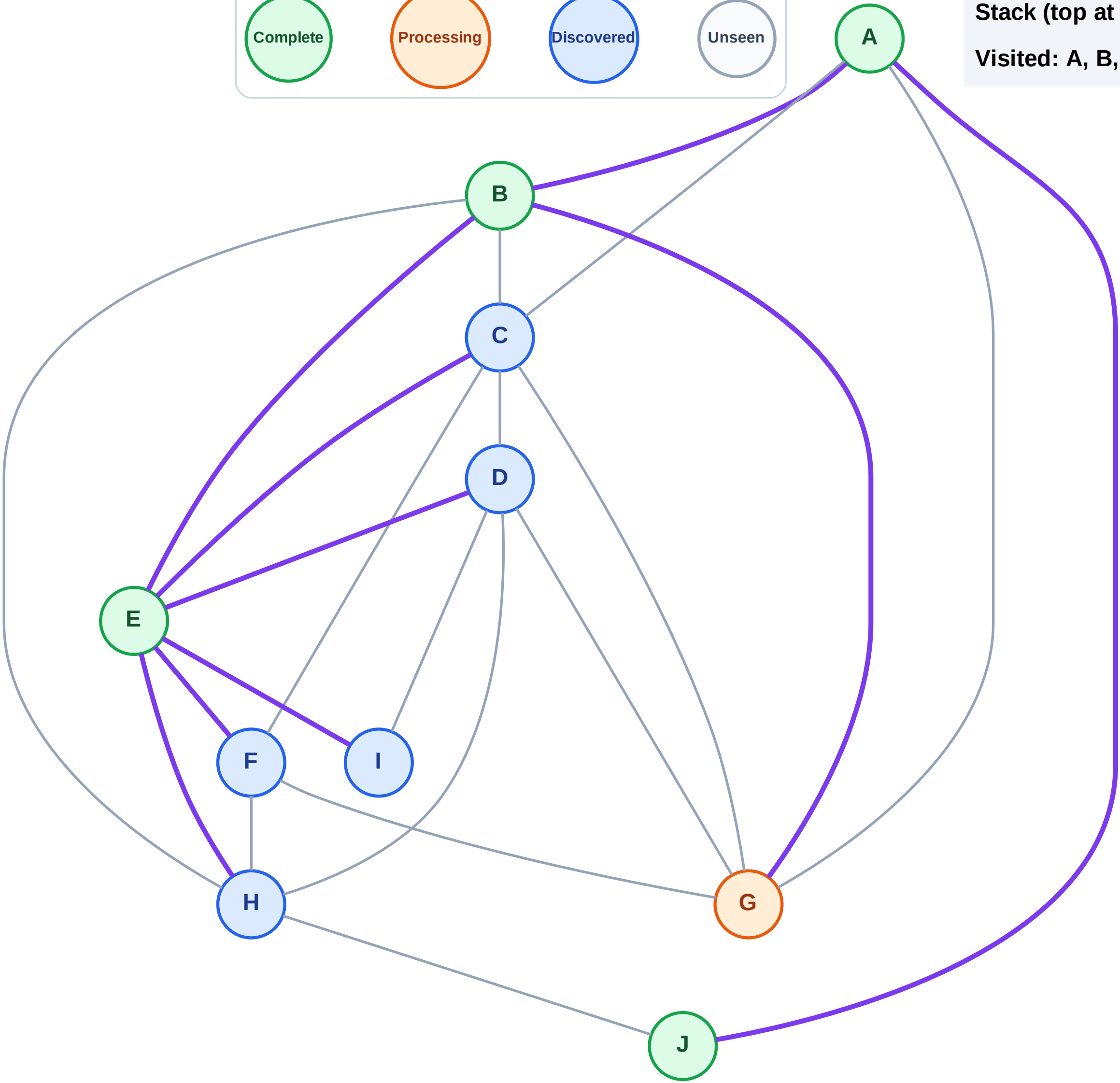
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C

Visited: A, B, E, J



DFS Traversal

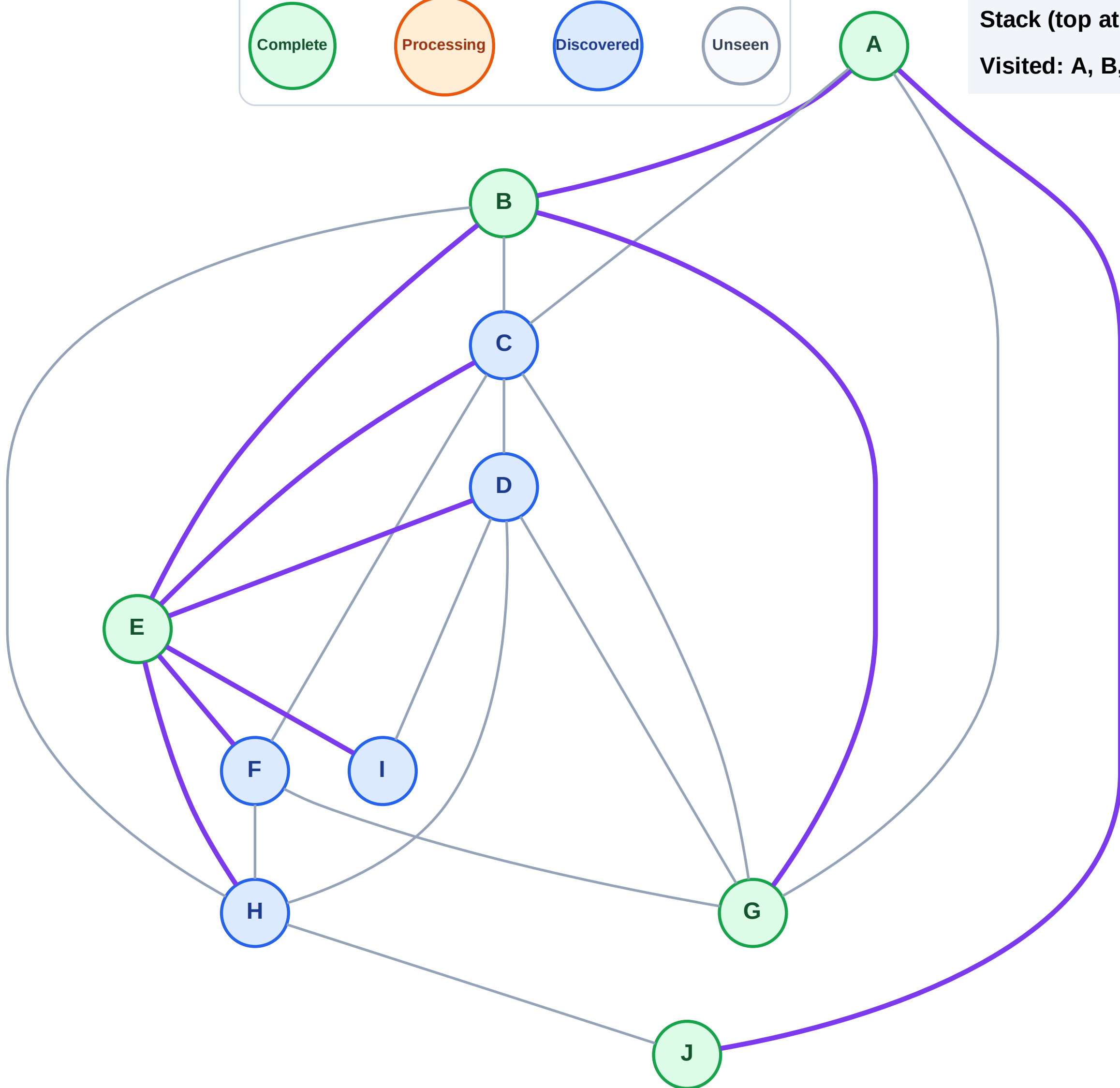
Step 11: No new neighbors from G

Legend



Stack (top at right): I | H | F | D | C

Visited: A, B, E, G, J



DFS Traversal

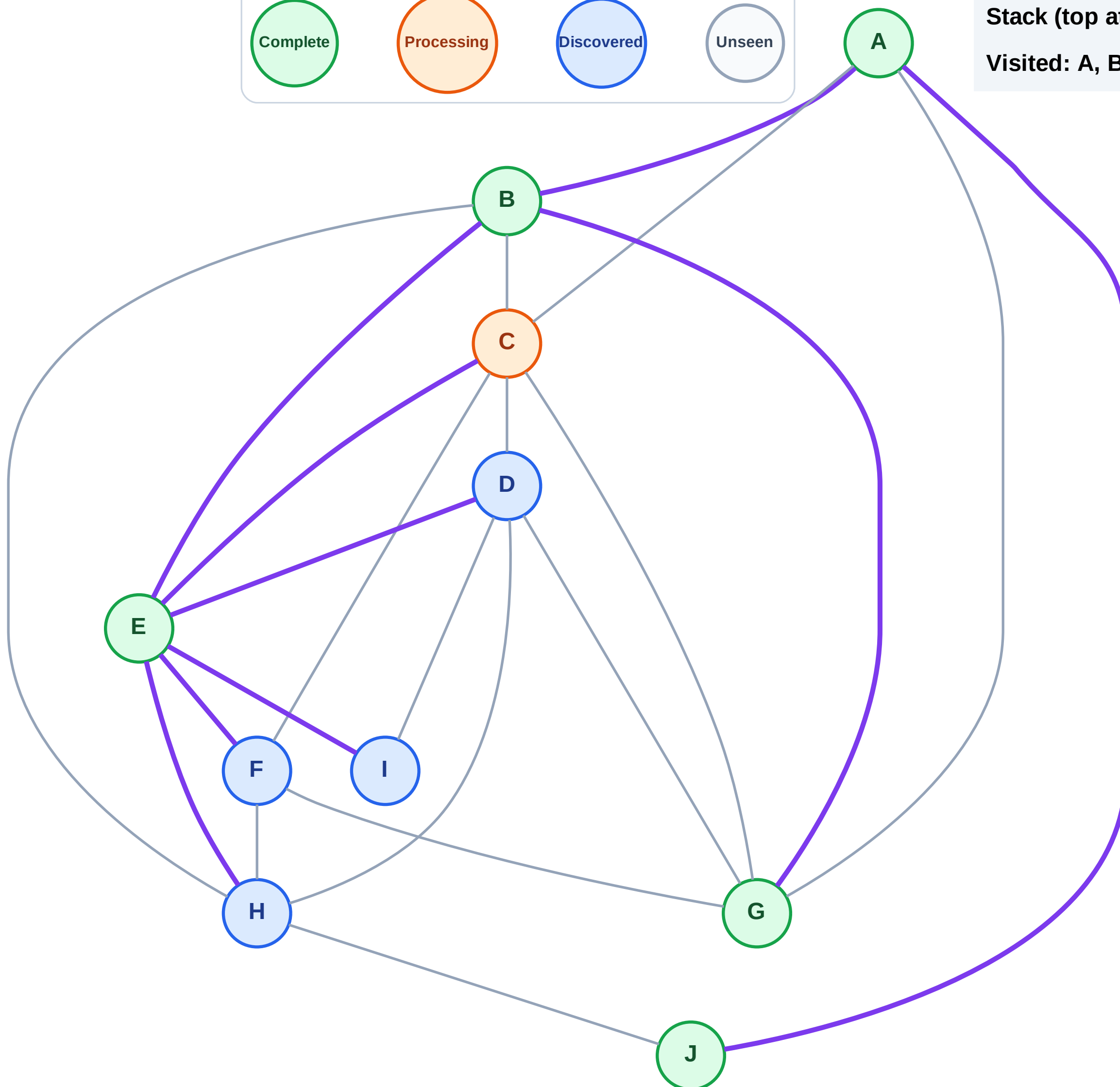
Step 12: Process node C

Legend



Stack (top at right): I | H | F | D

Visited: A, B, E, G, J



DFS Traversal

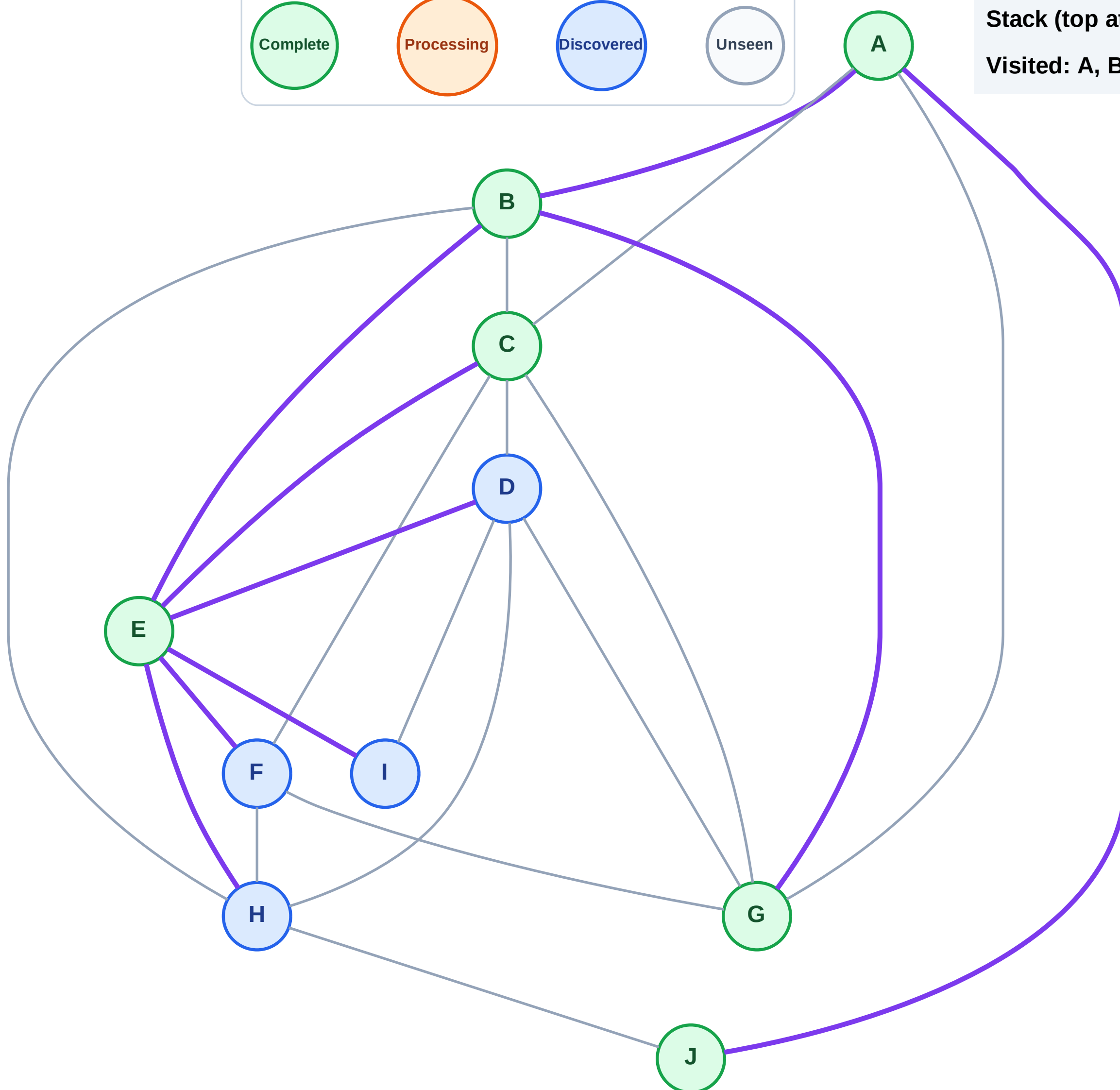
Step 13: No new neighbors from C

Legend



Stack (top at right): I | H | F | D

Visited: A, B, C, E, G, J



DFS Traversal

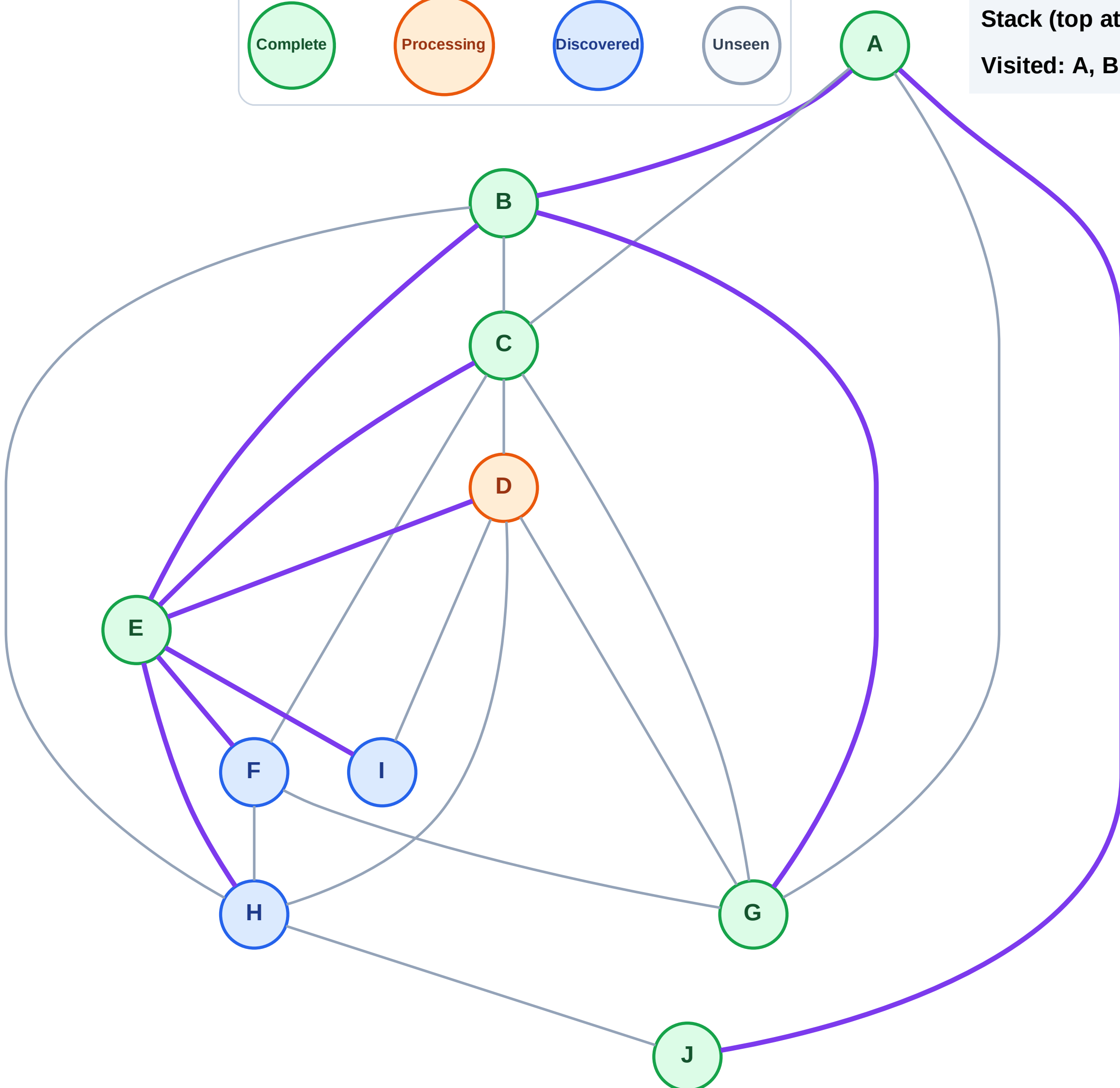
Step 14: Process node D

Legend



Stack (top at right): I | H | F

Visited: A, B, C, E, G, J



DFS Traversal

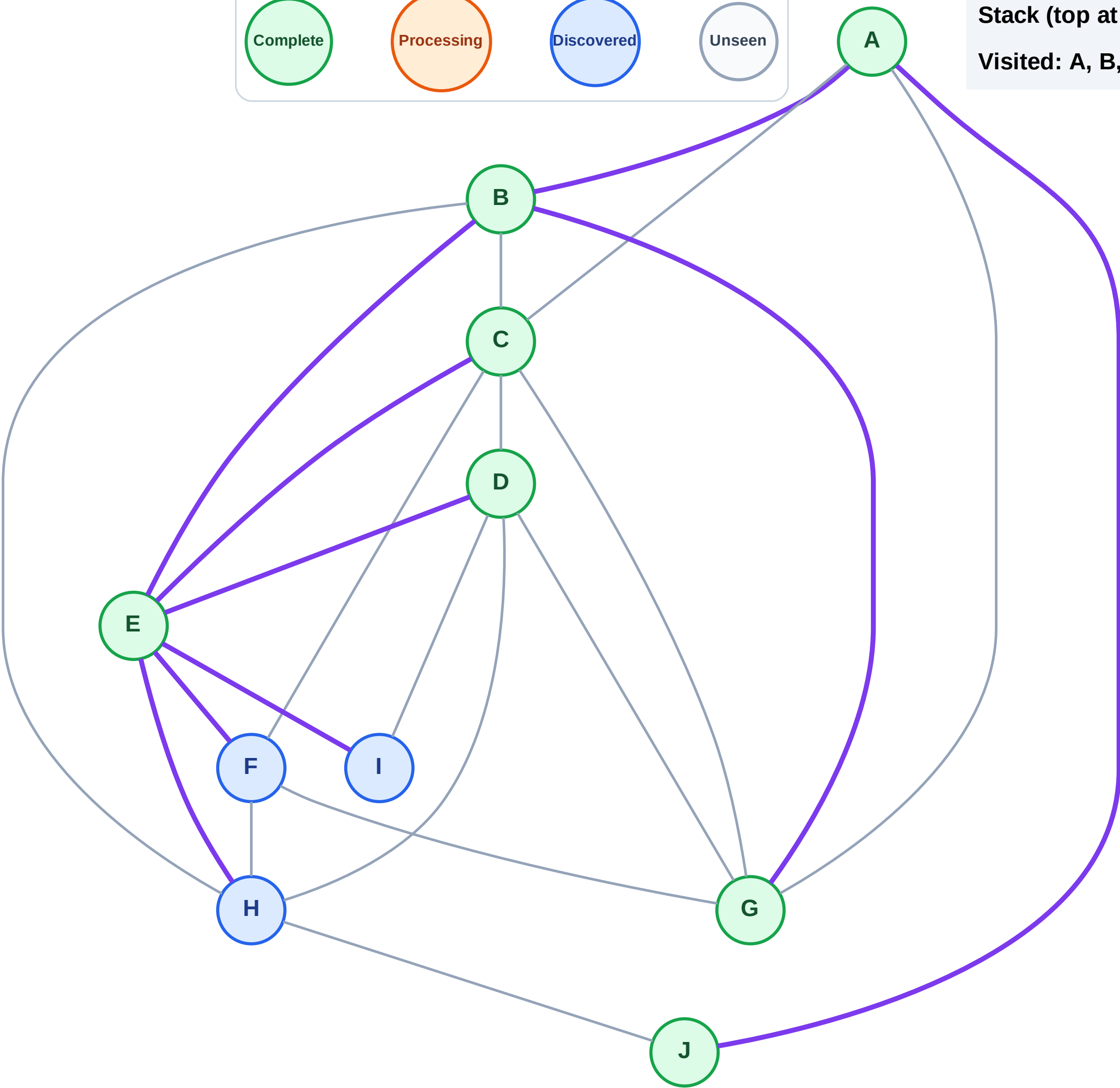
Step 15: No new neighbors from D

Legend



Stack (top at right): I | H | F

Visited: A, B, C, D, E, G, J



DFS Traversal

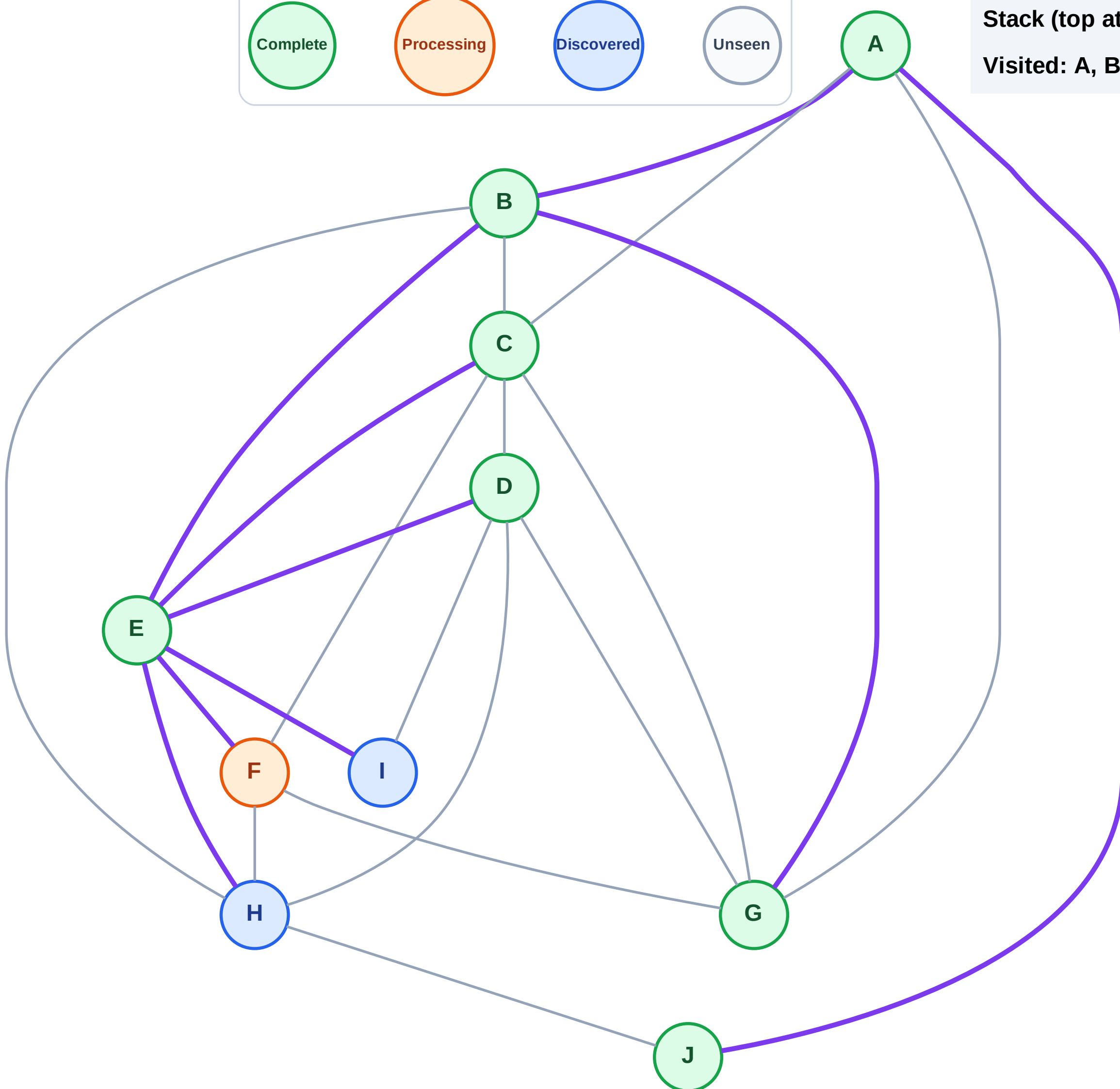
Step 16: Process node F

Legend



Stack (top at right): I | H

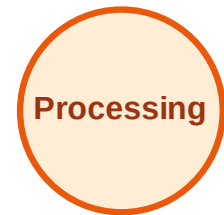
Visited: A, B, C, D, E, G, J



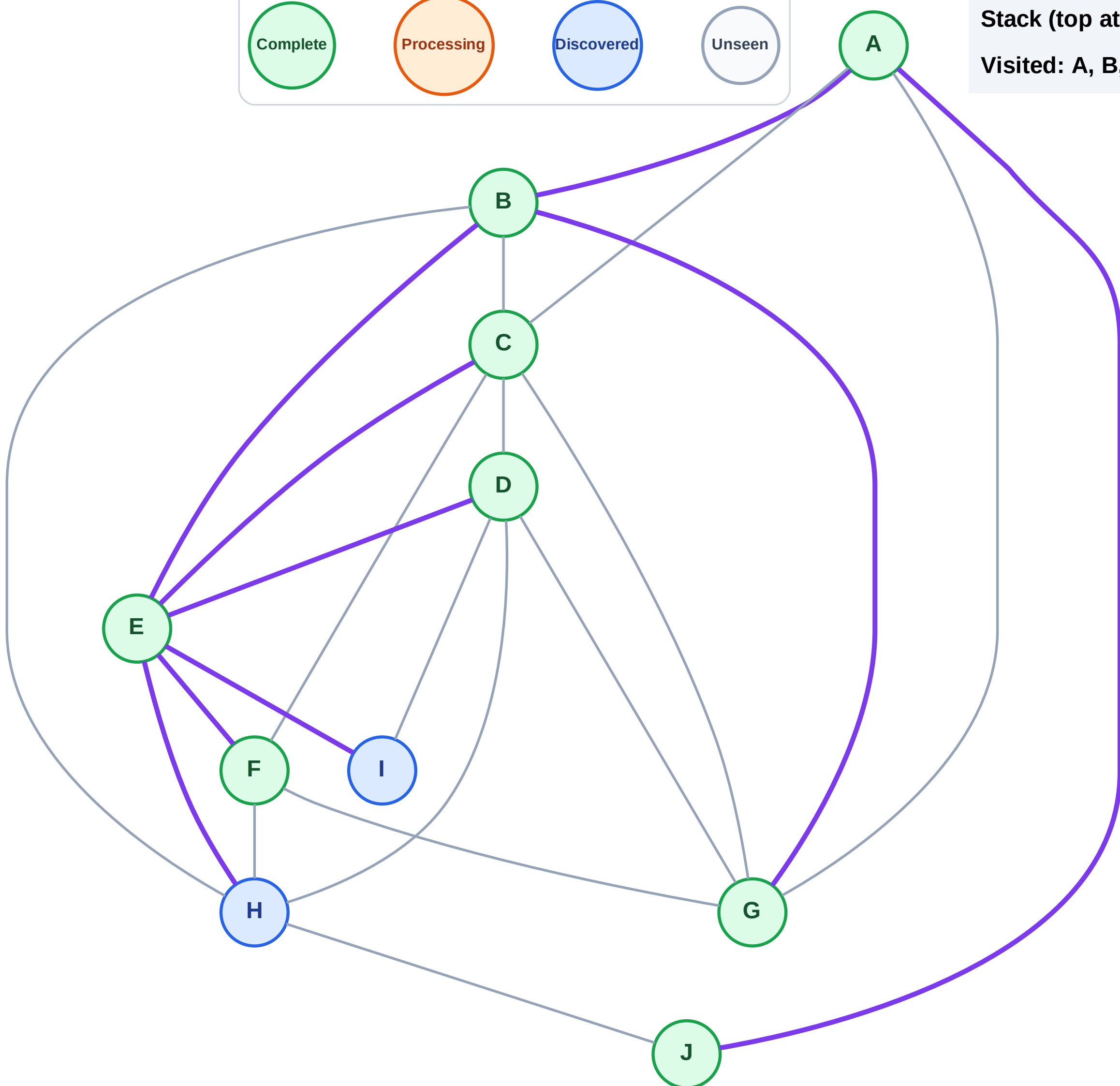
DFS Traversal

Step 17: No new neighbors from F

Legend



Stack (top at right): I | H
Visited: A, B, C, D, E, F, G, J



DFS Traversal

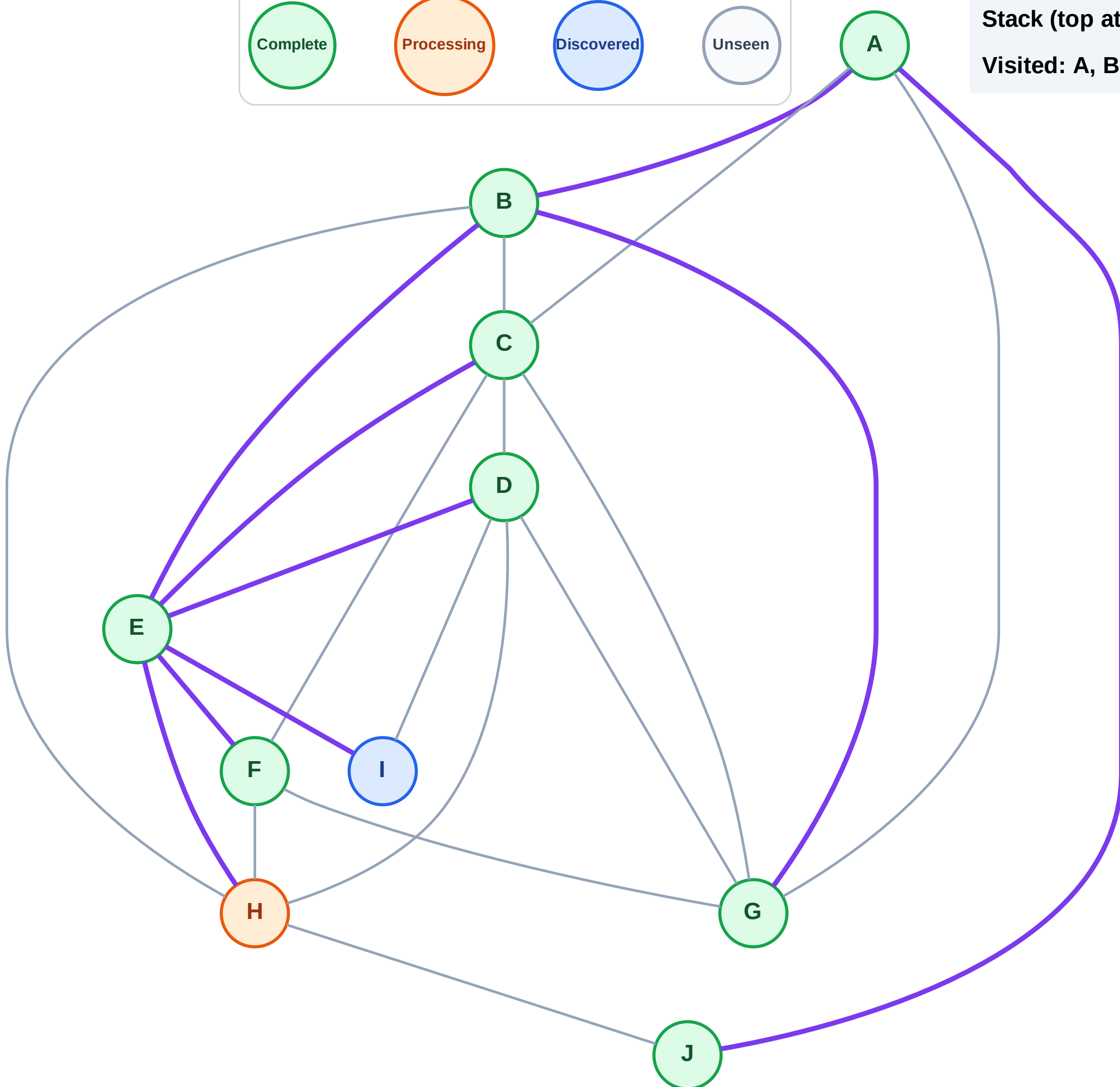
Step 18: Process node H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, J



DFS Traversal

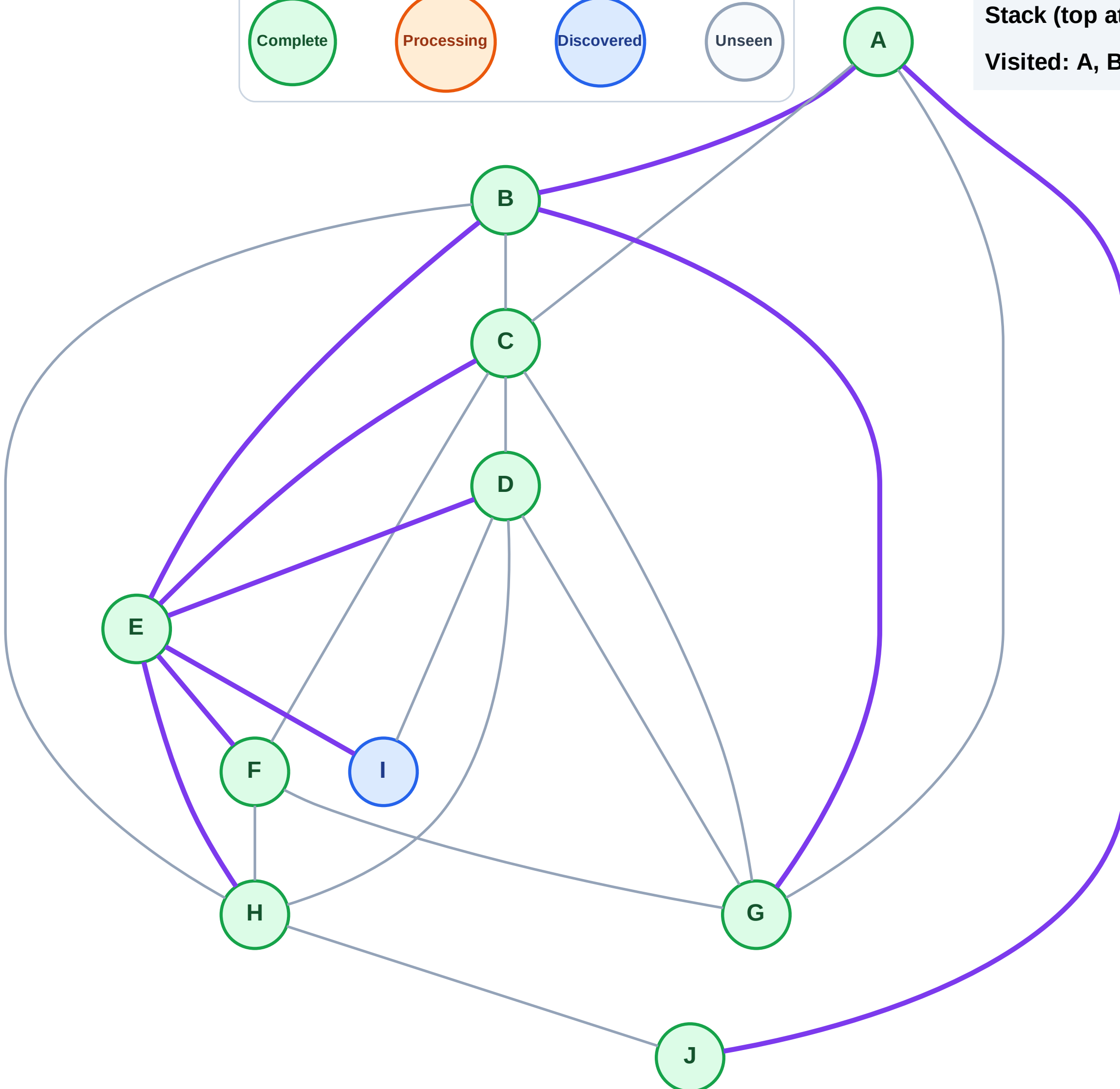
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

Complete

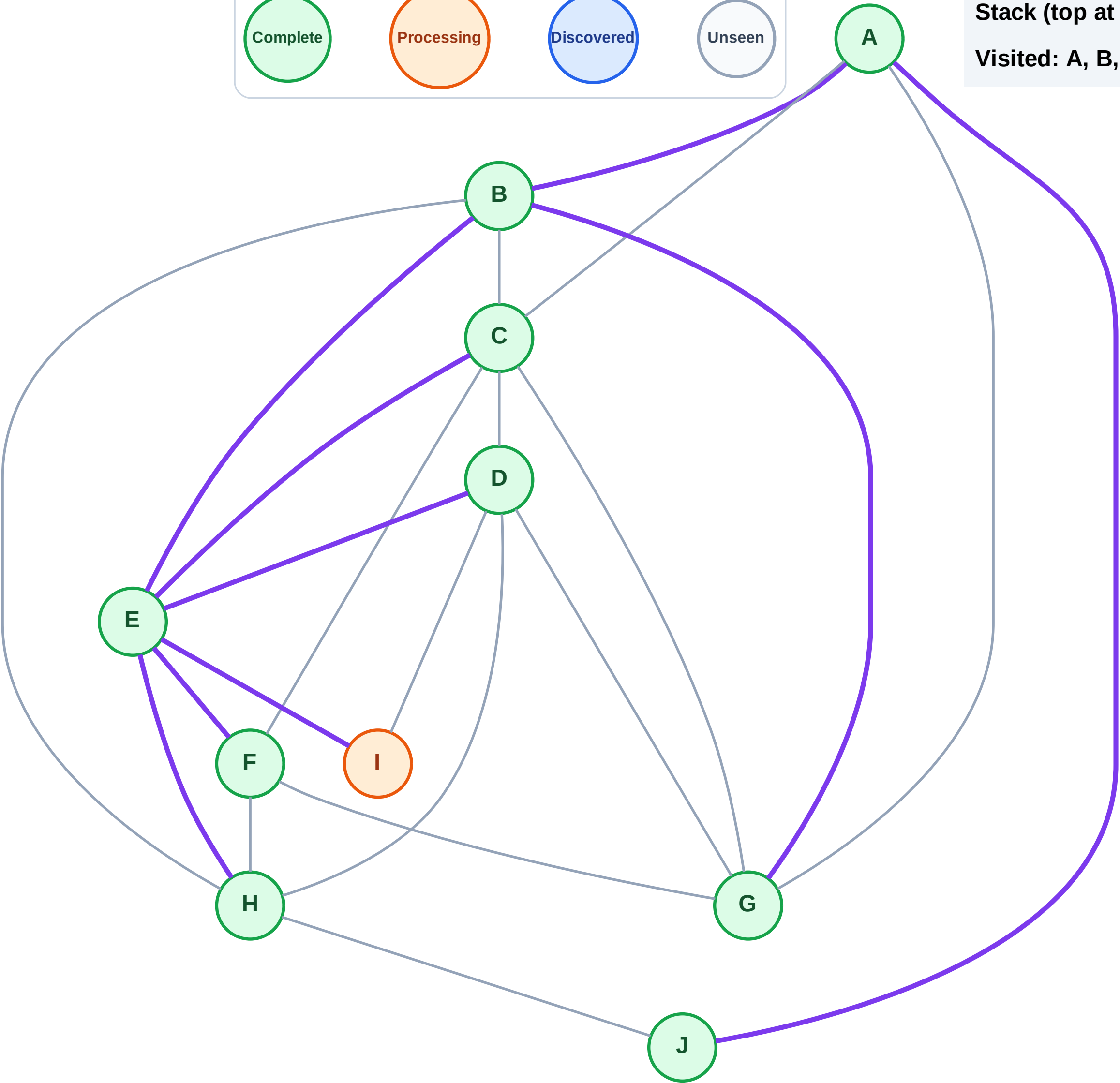
Processing

Discovered

Unseen

Stack (top at right): (empty)

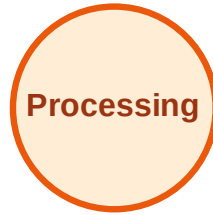
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

